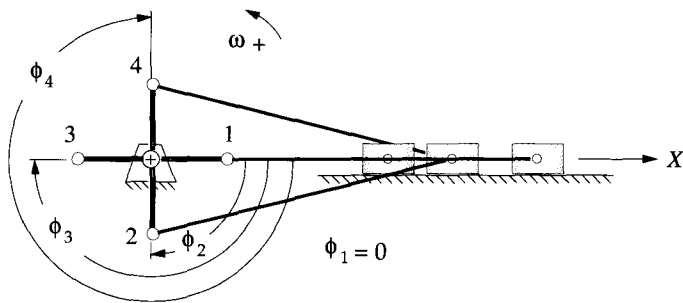
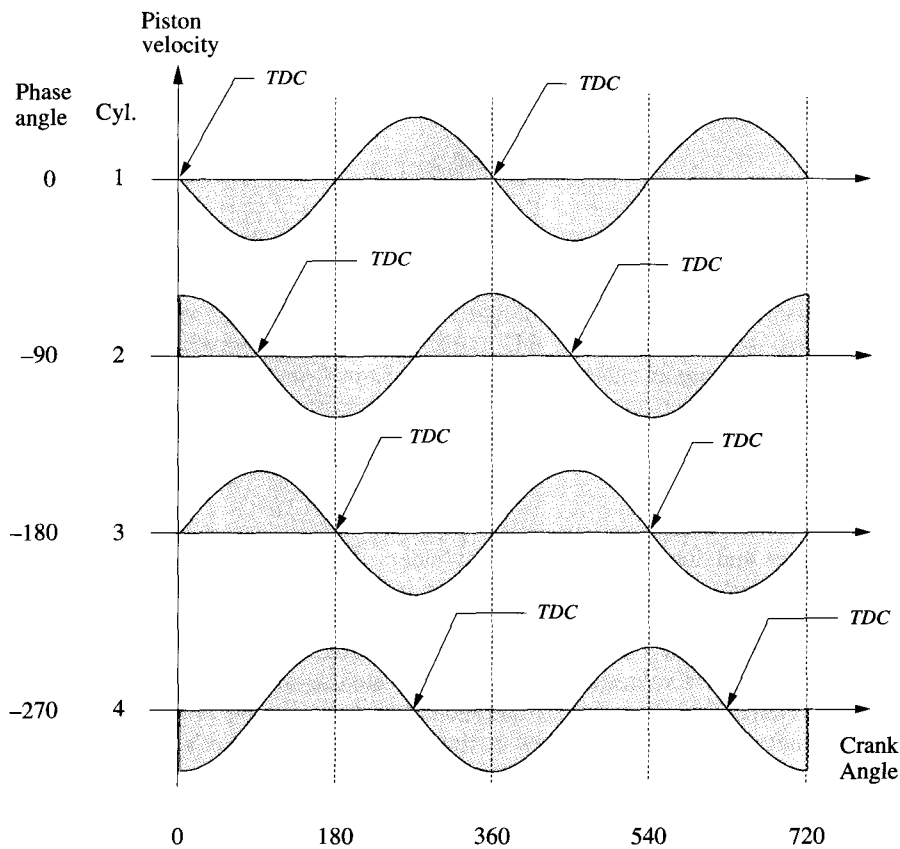




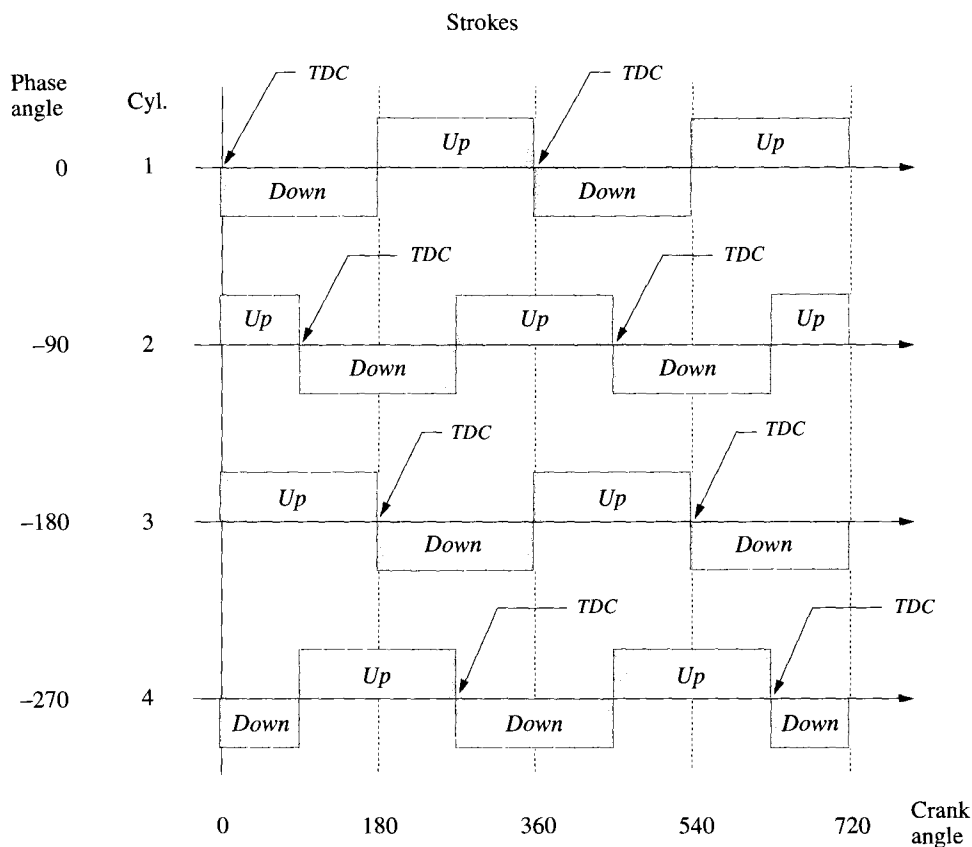
(a) Crankshaft phase angles



(b) The crank phase diagram

FIGURE 14-7

Crank phase angles and the phase diagram

**FIGURE 14-8**

The schematic crank phase diagram

cle of the ups and downs of the various cylinders. This same information is conveyed by the simplified crank phase diagram shown in Figure 14-8. Here the piston motions are represented by rectangular blocks with a negative block arbitrarily used to denote a piston downstroke and a positive one a piston upstroke. It is strictly schematic. The positive and negative values of the blocks imply nothing more than that stated. Such a schematic crank phase diagram can (and should) be drawn for any proposed arrangement of crankshaft phase angles. To draw it, simply shift each cylinder's blocks to the right by its phase angle with respect to the first cylinder.

14.3 SHAKING FORCES IN INLINE ENGINES

We want to determine the overall shaking force which results from our chosen crankshaft phase angle arrangement. The individual cylinders will each contribute to the total shaking force. We can superpose their effects, taking their phase shifts into account. Equa-

tion 13.14e (p. 619) defined the shaking force for one cylinder with the crankshaft rotating at constant ω .

for $\alpha = 0$:

$$\mathbf{F}_s \cong \left[m_A r \omega^2 \cos \omega t + m_B r \omega^2 \left(\cos \omega t + \frac{r}{l} \cos 2\omega t \right) \right] \hat{\mathbf{i}} + \left[m_A r \omega^2 \sin \omega t \right] \hat{\mathbf{j}} \quad (14.2a)$$

This expression is for an unbalanced crank. In multicylinder engines each crank throw on the crankshaft is at least counterweighted to eliminate the shaking force effects of the combined mass m_A of crank and conrod assumed concentrated at the crankpin. (See Section 13.10 and equation 13.24, p. 631.) Sometimes the crank throws in a multicylinder engine are also overbalanced, although to a lesser extent than for a one-cylinder engine.* The need for overbalancing is less if the crankshaft phase angles are arranged to cancel the effects of the reciprocating masses at the wrist pins. This is possible except in some two-cylinder, four-stroke, inline engines.

If we provide balance masses with an mr product equal to $m_A r_A$ on each crank throw as shown in Figure 14-3 (p. 641), the terms in equation 14.2a which include m_A will be eliminated, reducing it to:

$$\mathbf{F}_s \cong m_B r \omega^2 \left(\cos \omega t + \frac{r}{l} \cos 2\omega t \right) \hat{\mathbf{i}} \quad (14.2b)$$

Recall that these are approximate expressions which exclude all harmonics above the second.

We will assume that all cylinders in the engine are of equal displacement and that all pistons and all conrods are interchangeable. This is desirable both for dynamic balance and for lower production costs. If we let the crank angle ωt represent the instantaneous position of the reference crank throw for cylinder 1, the corresponding positions of the other cranks can be defined by their phase angles as shown in Figure 14-7 (p. 645). The total shaking force for a multicylinder inline engine is then:

$$\mathbf{F}_s \cong m_B r \omega^2 \sum_{i=1}^n \left[\cos(\omega t - \phi_i) + \frac{r}{l} \cos 2(\omega t - \phi_i) \right] \hat{\mathbf{i}} \quad (14.2c)$$

where n = number of cylinders and $\phi_1 = 0$. Substitute the identity:

$$\cos(a - b) = \cos a \cos b + \sin a \sin b$$

and factor to:

$$\mathbf{F}_s \cong m_B r \omega^2 \left[\cos \omega t \sum_{i=1}^n \cos \phi_i + \sin \omega t \sum_{i=1}^n \sin \phi_i + \frac{r}{l} \left(\cos 2\omega t \sum_{i=1}^n \cos 2\phi_i + \sin 2\omega t \sum_{i=1}^n \sin 2\phi_i \right) \right] \hat{\mathbf{i}} \quad (14.2d)$$

The ideal value for the shaking force is zero. This expression can only be zero for all values of ωt if:

* A 90° vee-eight engine typically has approximately $m_B/2$ of overbalance mass added per crank throw.

$$\sum_{i=1}^n \cos \phi_i = 0 \qquad \sum_{i=1}^n \sin \phi_i = 0 \qquad (14.3a)$$

$$\sum_{i=1}^n \cos 2\phi_i = 0 \qquad \sum_{i=1}^n \sin 2\phi_i = 0 \qquad (14.3b)$$

Thus, there are some combinations of phase angles ϕ_i which will cause cancellation of the shaking force through the second harmonic. If we wish to cancel higher harmonics, we could replace those harmonics' terms truncated from the Fourier series representation and find that the fourth and sixth harmonics will be cancelled if:

$$\sum_{i=1}^n \cos 4\phi_i = 0 \qquad \sum_{i=1}^n \sin 4\phi_i = 0 \qquad (14.3c)$$

$$\sum_{i=1}^n \cos 6\phi_i = 0 \qquad \sum_{i=1}^n \sin 6\phi_i = 0 \qquad (14.3d)$$

Equations 14.3 provide us with a convenient predictor of the shaking force behavior of any proposed inline engine design. Program ENGINE calculates equations 14.3 and displays a table of their values. Note that **both the sine and cosine summations of any multiple of the phase angles must be zero for that harmonic of the shaking force to be zero**. The calculation for our example of a four-cylinder engine with phase angles of $\phi_1 = 0$, $\phi_2 = 90$, $\phi_3 = 180$, $\phi_4 = 270^\circ$ in Table 14-1 shows that the shaking forces are zero for the first, second, and sixth harmonics and nonzero for the fourth. So, our common-sense choice in this instance has proven a good one as far as shaking forces are concerned. The coefficients of the fourth and sixth harmonic terms are minuscule, so their contributions, if any, can be ignored. The primary component is of most concern, because of its potential magnitude. The secondary (second harmonic) term is less critical than the primary as it is multiplied by r/l which is generally less than 1/3. An unbalanced secondary harmonic of shaking force is undesirable but can be lived with, especially if the engine is of small displacement (less than about 1/2 liter per cylinder).

TABLE 14-1 Force Balance State of a 4-Cylinder Inline Engine with a 0, 90, 180, 270° Crankshaft

Primary forces :	$\sum_{i=1}^n \sin \phi_i = 0$	$\sum_{i=1}^n \cos \phi_i = 0$
Secondary forces :	$\sum_{i=1}^n \sin 2\phi_i = 0$	$\sum_{i=1}^n \cos 2\phi_i = 0$
Fourth harmonic forces :	$\sum_{i=1}^n \sin 4\phi_i = 0$	$\sum_{i=1}^n \cos 4\phi_i = 4$
Sixth harmonic forces :	$\sum_{i=1}^n \sin 6\phi_i = 0$	$\sum_{i=1}^n \cos 6\phi_i = 0$

To see more detail on the results of this 0, 90, 180, 270 inline four-cylinder engine configuration, run program ENGINE, select the configuration from the *Example* pull-down menu, and then *Plot* the shaking force. See Appendix A for more detailed instructions on the use of program ENGINE.

14.4 INERTIA TORQUE IN INLINE ENGINES

The inertia torque for a single-cylinder engine was defined in Section 13.6 and equation 13.15e (p. 612). We are concerned with reducing this inertia torque, preferably to zero, because it adds to the gas torque to form the total torque. (See Section 13.7, p. 622.) Inertia torque adds nothing to the net driving torque as its average value is always zero, but it does create large oscillations in the total torque which detracts from its smoothness. Inertia torque oscillations can be masked to a degree with the addition of sufficient flywheel to the system, or their external, net effect can be cancelled by the proper choice of phase angles. However, the torque oscillations, even if hidden from the outside observer, or made to sum to zero, are still present within the crankshaft and can lead to torsional fatigue failure if the part is not properly designed. (See also Figure 14-22, p. 671.) The approximate one-cylinder inertia torque equation for three harmonics is:

$$\mathbf{T}_{i21} \cong \frac{1}{2} m_B r^2 \omega^2 \left(\frac{r}{2l} \sin \omega t - \sin 2\omega t - \frac{3r}{2l} \sin 3\omega t \right) \hat{\mathbf{k}} \quad (14.4a)$$

Summing for all cylinders and including their phase angles:

$$\mathbf{T}_{i21} \cong \frac{1}{2} m_B r^2 \omega^2 \sum_{i=1}^n \left[\frac{r}{2l} \sin(\omega t - \phi_i) - \sin 2(\omega t - \phi_i) - \frac{3r}{2l} \sin 3(\omega t - \phi_i) \right] \hat{\mathbf{k}} \quad (14.4b)$$

Substitute the identity:

$$\sin(a - b) = \sin a \cos b - \cos a \sin b$$

and factor to:

$$\mathbf{T}_{i21} \cong \frac{1}{2} m_B r^2 \omega^2 \left[\begin{aligned} & \frac{r}{2l} \left(\sin \omega t \sum_{i=1}^n \cos \phi_i - \cos \omega t \sum_{i=1}^n \sin \phi_i \right) \\ & - \left(\sin 2\omega t \sum_{i=1}^n \cos 2\phi_i - \cos 2\omega t \sum_{i=1}^n \sin 2\phi_i \right) \\ & - \frac{3r}{2l} \left(\sin 3\omega t \sum_{i=1}^n \cos 3\phi_i - \cos 3\omega t \sum_{i=1}^n \sin 3\phi_i \right) \end{aligned} \right] \hat{\mathbf{k}} \quad (14.4c)$$

This can only be zero for all values of ωt if:

$$\sum_{i=1}^n \sin \phi_i = 0 \qquad \sum_{i=1}^n \cos \phi_i = 0 \qquad (14.5a)$$

$$\sum_{i=1}^n \sin 2\phi_i = 0 \qquad \sum_{i=1}^n \cos 2\phi_i = 0 \qquad (14.5b)$$

$$\sum_{i=1}^n \sin 3\phi_i = 0 \qquad \sum_{i=1}^n \cos 3\phi_i = 0 \qquad (14.5c)$$

Equations 14.5 provide us with a convenient predictor of the inertia torque behavior of any proposed inline engine design. Calculation for our example of a four-cylinder engine with phase angles of $\phi_1 = 0$, $\phi_2 = 90$, $\phi_3 = 180$, $\phi_4 = 270^\circ$ shows that the inertia torque components are zero for the first, second, and third harmonics. So, our current example is a good one for inertia torques as well.

14.5 SHAKING MOMENT IN INLINE ENGINES

We were able to consider the single-cylinder engine to be a single plane, or two dimensional, device and thus could statically balance it. The multicylinder engine is three dimensional. Its multiple cylinders are distributed along the axis of the crankshaft. Even though we may have cancellation of the shaking forces, there may still be unbalanced moments in the plane of the engine block. We need to apply the criteria for dynamic balance. (See Section 12.2 and equation 12.3, p. 574.) Figure 14-9 shows a schematic of an inline four-cylinder engine with crank phase angles of $\phi_1 = 0$, $\phi_2 = 90$, $\phi_3 = 180$, $\phi_4 = 270^\circ$. The spacing between the cylinders is normally uniform. We can sum moments in the plane of the cylinders about any convenient point such as one end of the crankshaft,

$$\sum M_L = \sum_{i=1}^n z_i \mathbf{F}_{s_i} \hat{\mathbf{j}} \qquad (14.6a)$$

where $\mathbf{F}_{s_i}$ is the shaking force and z_i is the moment arm of the i th cylinder. Substituting equation 14.2d for $\mathbf{F}_{s_i}$:

$$\sum M_L \cong m_B r \omega^2 \left[\begin{aligned} &\cos \omega t \sum_{i=1}^n z_i \cos \phi_i + \sin \omega t \sum_{i=1}^n z_i \sin \phi_i \\ &+ \frac{r}{l} \left(\cos 2\omega t \sum_{i=1}^n z_i \cos 2\phi_i + \sin 2\omega t \sum_{i=1}^n z_i \sin 2\phi_i \right) \end{aligned} \right] \hat{\mathbf{j}} \qquad (14.6b)$$

This expression can only be zero for all values of ωt if:

$$\sum_{i=1}^n z_i \cos \phi_i = 0 \qquad \sum_{i=1}^n z_i \sin \phi_i = 0 \qquad (14.7a)$$

$$\sum_{i=1}^n z_i \cos 2\phi_i = 0 \qquad \sum_{i=1}^n z_i \sin 2\phi_i = 0 \qquad (14.7b)$$

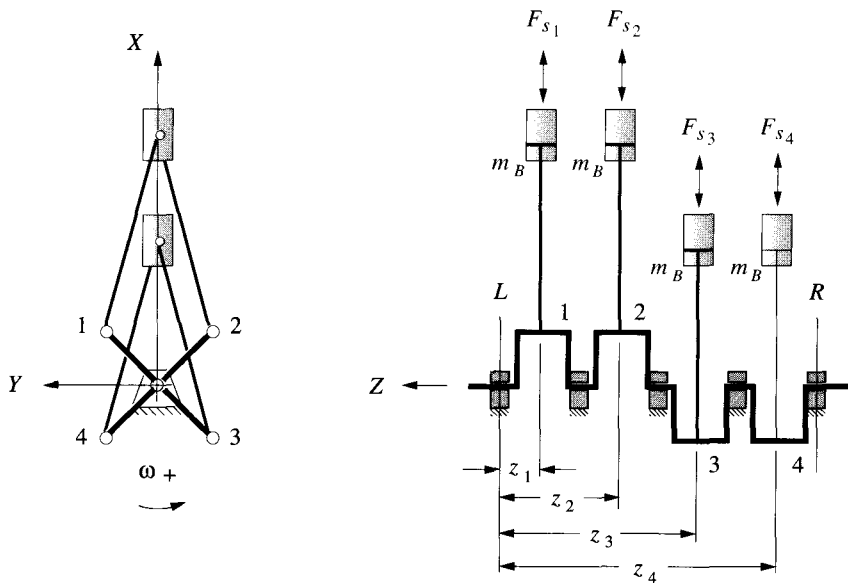


FIGURE 14-9

Moment arms of the shaking moment

These will guarantee no shaking moments through the second harmonic. We can extend this to higher harmonics as we did for the shaking force.

$$\sum_{i=1}^n z_i \cos 4\phi_i = 0 \quad \sum_{i=1}^n z_i \sin 4\phi_i = 0 \quad (14.7c)$$

$$\sum_{i=1}^n z_i \cos 6\phi_i = 0 \quad \sum_{i=1}^n z_i \sin 6\phi_i = 0 \quad (14.7d)$$

Note that both the sine and cosine summations of any multiple of the phase angles must be zero for that harmonic of the shaking moment to be zero. The calculation for our example of a four-cylinder engine with phase angles of $\phi_1 = 0$, $\phi_2 = 90$, $\phi_3 = 180$, $\phi_4 = 270^\circ$, and an assumed cylinder spacing of one length unit ($z_1 = 1$) in Table 14-2, shows that the shaking moments are not zero for any of these harmonics. So, our choice of phase angles which is a good one for shaking forces and torques fails the test for zero shaking moments. The coefficients of the fourth and sixth harmonic terms in the moment equations are minuscule, so they can be ignored. The secondary (second harmonic) term is less critical than the primary as it is multiplied by r/l which is generally less than $1/3$. An unbalanced secondary harmonic of shaking moment is undesirable but can be tolerated, especially if the engine is of small displacement (less than about $1/2$ liter per cylinder). The primary component is of most concern, because of its magnitude. If we wish to use this crankshaft configuration, we will need to apply a balancing technique to the engine as described in a later section to at least eliminate the primary moment. A large shaking moment is undesirable as it will cause the engine to **pitch** forward and back

TABLE 14-2 Moment Balance State of a 4-Cylinder, Inline Engine with a 0, 90, 180, 270° Crankshaft, and $z_1 = 1$, $z_2 = 2$, $z_3 = 3$, $z_4 = 4$

Primary moments :	$\sum_{i=1}^n z_i \sin \phi_i = -2$	$\sum_{i=1}^n z_i \cos \phi_i = -2$
Secondary moments :	$\sum_{i=1}^n z_i \sin 2\phi_i = 0$	$\sum_{i=1}^n z_i \cos 2\phi_i = -2$
Fourth harmonic moments :	$\sum_{i=1}^n z_i \sin 4\phi_i = 0$	$\sum_{i=1}^n z_i \cos 4\phi_i = 6$
Sixth harmonic moments :	$\sum_{i=1}^n z_i \sin 6\phi_i = 0$	$\sum_{i=1}^n z_i \cos 6\phi_i = -2$

(like a bucking bronco) as the moment oscillates from positive to negative in the plane of the cylinders. *Do not confuse this shaking moment with the shaking torque* which acts to **roll** the engine back and forth about the axis of the crankshaft.

Figure 14-10 shows the primary and secondary components of the shaking moment for this example engine for two revolutions of the crank. Each is a pure harmonic of zero average value. The total moment is the sum of these two components. Use example engine #6 in program ENGINE to view these functions. See Appendix A for help with the program.

14.6 EVEN FIRING

The inertial forces, torques, and moments are only one set of criteria which need to be considered in the design of multicylinder engines. Gas force and gas torque considerations are equally important. In general, it is desirable to create a firing pattern among the cylinders that is evenly spaced in time. If the cylinders fire unevenly, vibrations will be created which may be unacceptable. Smoothness of the power pulses is desired. The power pulses depend on the stroke cycle. If the engine is a two-stroke, there will be one power pulse per revolution in each of its n cylinders. The optimum delta phase angle between the cylinders' crank throws for evenly spaced power pulses will then be:

$$\Delta\phi_{two\ stroke} = \frac{360^\circ}{n} \quad (14.8a)$$

For a four-stroke engine there will be one power pulse in each cylinder every two revolutions. The optimum delta phase angle of the crank throws for evenly spaced power pulses will then be:

$$\Delta\phi_{four\ stroke} = \frac{720^\circ}{n} \quad (14.8b)$$

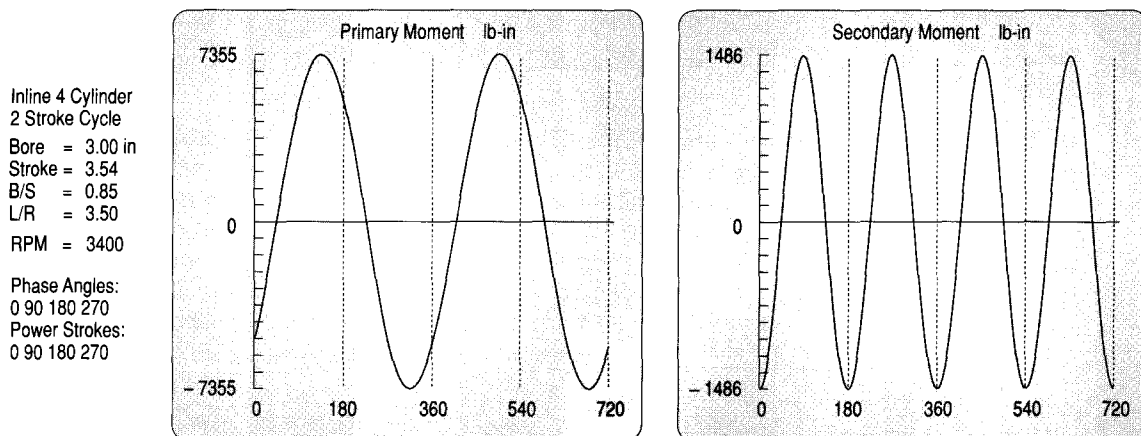


FIGURE 14-10

Primary and secondary moments in the 0, 90, 180, 270° crankshaft four-cylinder engine

Compare equations 14.8a and 14.8b to equation 14.1 (p. 644) which defined the optimum delta phase angle for cancellation of inertia forces. A two-stroke engine can have both even firing and inertia balance, but a four-stroke engine has a conflict between these two criteria. Thus some design trade-offs will be necessary to obtain the best compromise between these factors in the four-stroke case.

Two-Stroke Cycle Engine

To determine the firing pattern of an engine design we must return to the crank phase diagram. Figure 14-11 reproduces Figure 14-8 and adds new information to it. It shows the power pulses for a **two-stroke cycle, four-cylinder engine** with the $\phi_i = 0, 90, 180, 270^\circ$ phase angle crank configuration. Note that each cylinder's negative block in Figure 14-11 is shifted to the right by its phase angle with respect to the reference cylinder 1. In this schematic representation, only the negative blocks on the diagram are available for power pulses as they represent the downstroke of the piston. By convention, cylinder one fires first, so its negative block at 0° is labeled **Power**. The other cylinders may be fired in any order, but their power pulses should be as evenly spaced as possible across the interval.

The available power pulse spacings are dictated by the crank phase angles. There may be more than one firing order which will give even firing, especially with large numbers of cylinders. In this simple example the firing order 1, 2, 3, 4 will work as it will provide successive power pulses every 90° across the interval. The **power stroke angles** ψ_i are the angles in the cycle at which the cylinders fire. They are defined by the crankshaft phase angles and the choice of firing order in combination and in this example are $\psi_i = 0, 90, 180$, and 270° . In general, ψ_i are not equal to ϕ_i . Their correspondence with the phase angles in this example results from choosing the consecutive firing order 1, 2, 3, 4.

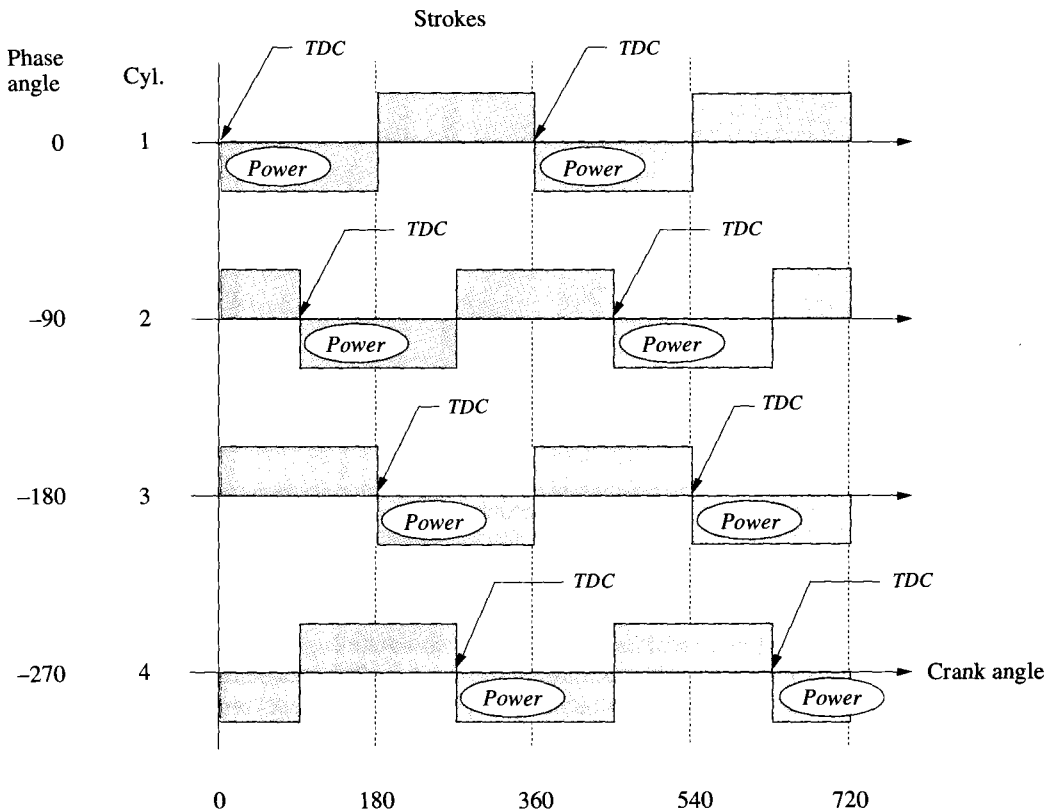


FIGURE 14-11

Two-stroke inline four-cylinder engine crank phase diagram with 0, 90, 180, 270° crankshaft phase angles

For a **two-stroke engine**, the power stroke angles ψ_i must be *between 0 and 360°*. We always want them to be evenly spaced in that interval with a delta power stroke angle defined by equation 14.8c. For our four-cylinder, two-stroke engine, ideal power stroke angles are then $\psi_i = 0, 90, 180, 270^\circ$, which we have achieved in this example.

We define the **delta power stroke angle** differently for each stroke cycle. For the two-stroke engine:

$$\Delta\psi_{two\ stroke} = \frac{360^\circ}{n} \quad (14.8c)$$

For the four-stroke engine:

$$\Delta\psi_{four\ stroke} = \frac{720^\circ}{n} \quad (14.8d)$$

The gas torque for a one-cylinder engine was defined in equation 13.8b (p. 613). The combined gas torque for all cylinders must sum the contributions of n cylinders, each phase-shifted by its power stroke angle ψ_i :

$$\mathbf{T}_{g21} \approx F_g r \sum_{i=1}^n \left\{ \sin(\omega t - \psi_i) \left[1 + \frac{r}{l} \cos(\omega t - \psi_i) \right] \right\} \hat{\mathbf{k}} \quad (14.9)$$

Figure 14-12 shows the gas torque, inertia torque, and shaking force for this two-stroke four-cylinder engine plotted from program ENGINE. The shaking moment components are shown in Figure 14-10 (p. 653). Except for the unbalanced shaking moments, this design is otherwise acceptable. The inertia force and inertia torque are both zero which is ideal. The gas torque consists of uniformly shaped and spaced pulses across the interval, four per revolution. Note that program ENGINE plots two full revolutions to accommodate the four-stroke case; thus eight power pulses are seen. Open the file F14-12.eng in the program to exercise this example.

Four-Stroke Cycle Engine

Figure 14-13 shows a crank phase diagram for the *same crankshaft design* as in Figure 14-11 except that it is designed as a *four-stroke cycle engine*. There is now only one power stroke every 720° for each cylinder. The second negative block for each cylinder must be used for the intake stroke. Cylinder 1 is again fired first. An evenly spaced pattern of power pulses among the other cylinders is again desired but is now not possible with this crankshaft. Whether the firing order is 1, 3, 4, 2 or 1, 2, 4, 3 or 1, 4, 2, 3, or any other chosen, there will be both gaps and overlaps in the power pulses. The first firing order listed, 1, 3, 4, 2, has been chosen for this example. This results in the set of power stroke angles $\psi_i = 0, 180, 270, 450^\circ$. These **power stroke angles** define the points in the **720° cycle** where each cylinder fires. Thus for a four-stroke engine, the power stroke angles ψ_i must be between 0 and 720° . We would like them to be evenly spaced in that interval with a delta angle defined by equation 14.8d. For our four-cylinder, four-stroke engine, the ideal power stroke angles would then be $\psi_i = 0, 180, 360, 540^\circ$. We clearly have not achieved them in this example. Figure 14-14 shows the resulting gas torque. Open the file F14-14.eng in program ENGINE to exercise this example.

The uneven firing in Figure 14-14 is obvious. This uneven gas torque will be perceived by the operator of any vehicle containing this engine as rough running and vibration, especially at idle speed. At higher engine speeds the flywheel will tend to mask this roughness, but flywheels are ineffective at low speeds. It is this fact that causes most engine designers to *favor even firing over elimination of inertia effects* in their selection of crankshaft phase angles. The inertia force, torque, and moment are all functions of engine speed squared. But, as engine speed increases the magnitude of these factors, the same speed is also increasing the flywheel's ability to mask their effects. Not so with gas-torque roughness due to uneven firing. It is bad at all speeds and the flywheel won't hide it at low speed.

We therefore must reject this crankshaft design for our four-stroke, four-cylinder engine. Equation 14.8b (p. 652) indicates that we need a delta phase angle $\Delta\phi_i = 180^\circ$ in our crankshaft to obtain even firing. We need four crank throws, and all crank phase angles must be less than 360° . So, we must repeat some angles if we use a delta phase angle of 180° . One possibility is $\phi_i = 0, 180, 0, 180^\circ$ for the four crank throws.* The crank phase diagram for this design is shown in Figure 14-15. The power strokes can

* Note that 0, 180, 360, 540, modulo 360 is the same as 0, 180, 0, 180.

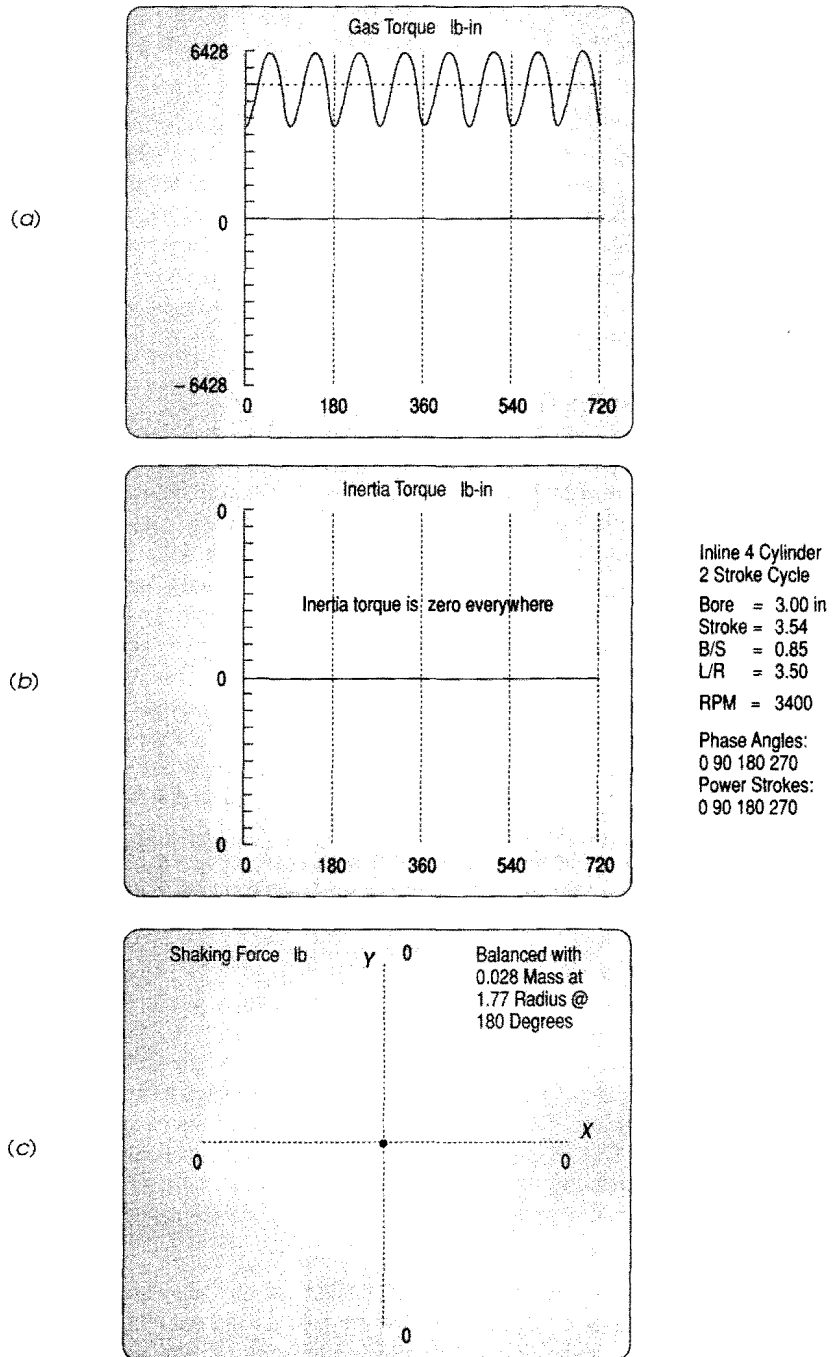


FIGURE 14-12

Torque and shaking force in the two-stroke four-cylinder inline engine

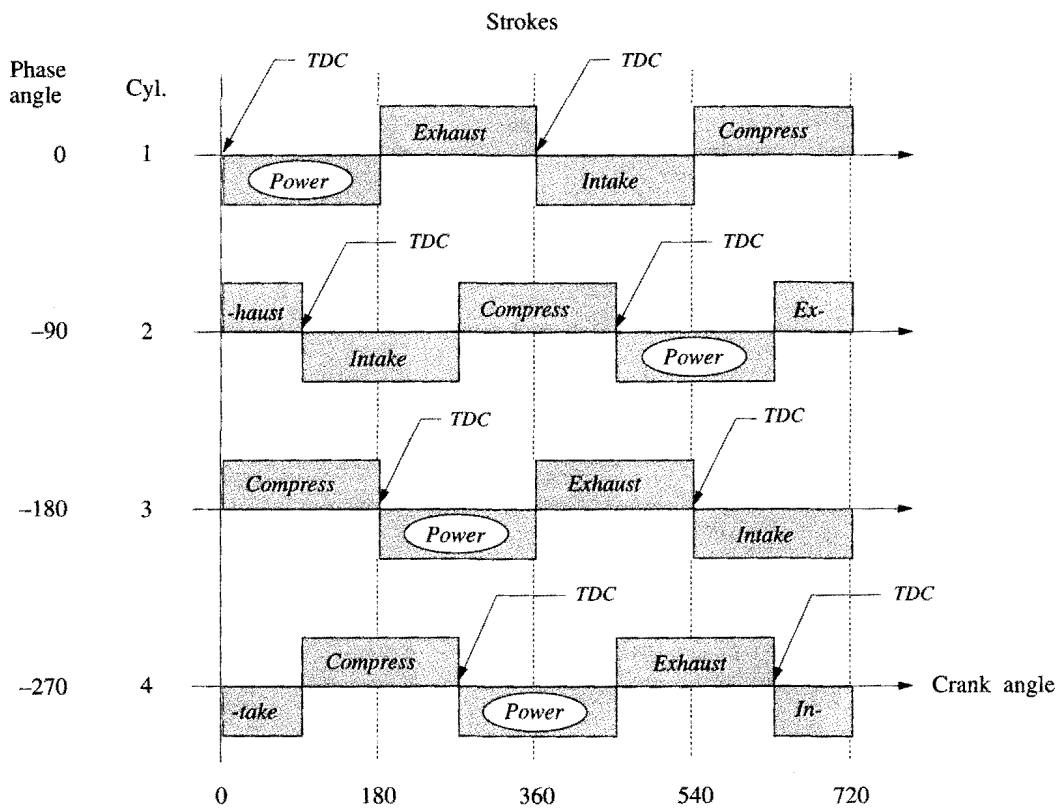


FIGURE 14-13
Four-stroke inline four-cylinder engine crank phase diagram with a 0, 90, 180, 270° crankshaft phase angles

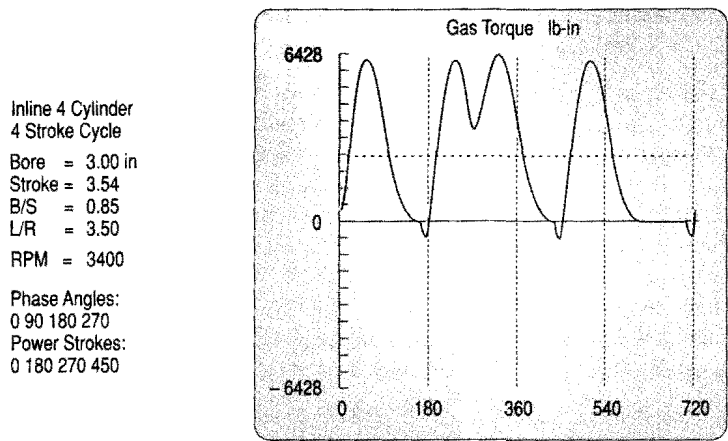


FIGURE 14-14
Uneven firing four-stroke, four-cylinder inline engine with a 0, 90, 180, 270° crankshaft

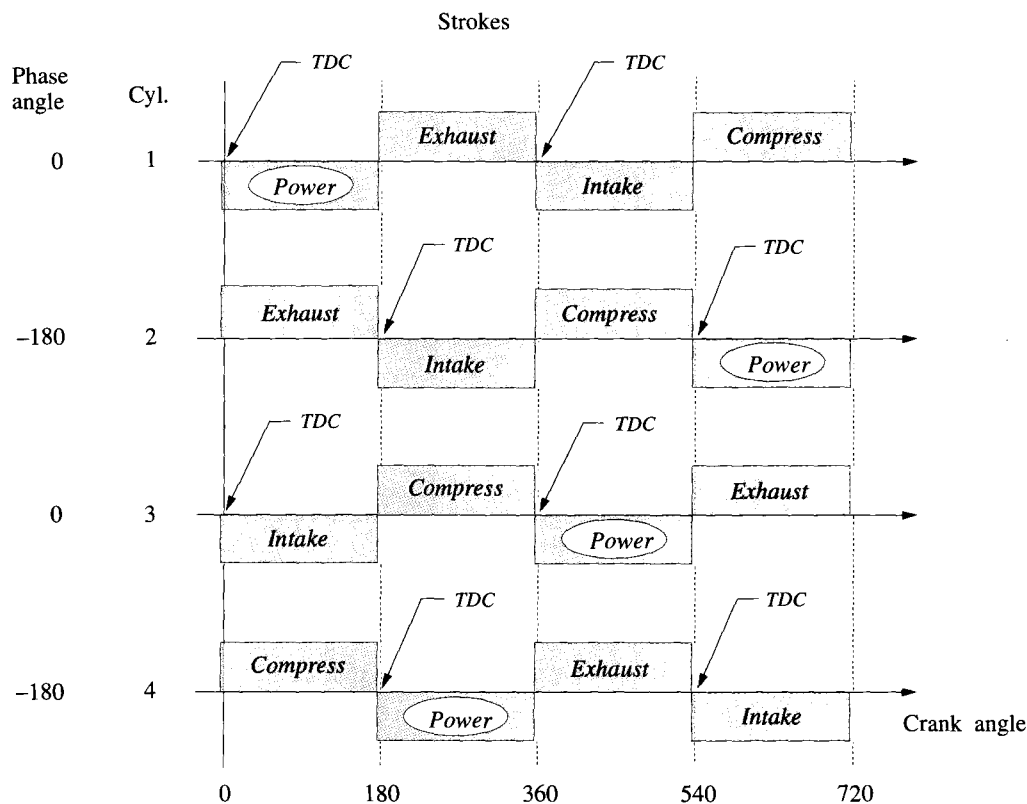


FIGURE 14-15

Even firing four-stroke, four-cylinder engine crank phase diagram with 0, 180, 0, 180° crankshaft phase angles

now be evenly spaced over 720°. A firing order of 1, 4, 3, 2 has been chosen which gives the desired sequence of power stroke angles, $\psi_i = 0, 180, 360, 540^\circ$. (Note that a firing order of 1, 2, 3, 4 would also work with this engine.)*

The inertial balance condition of this design must now be checked with equations 14.3, 14.5 and 14.7 (pp. 648-650). These show that the primary inertia force is zero, but the primary moment, secondary force, secondary moment, and inertia torque are all non-zero as shown in Table 14-3. So, this even-firing design has compromised the very good state of inertia balance of the previous design in order to achieve even firing. The inertia torque variations can be masked by a flywheel. The secondary forces and moments are relatively small in a small engine and can be tolerated. The nonzero primary moment is a problem which needs to be addressed. To see the results of this engine configuration, run program ENGINE and select it from the *Example* pull-down menu. Then plot the results. See Appendix A for more detailed instructions on the use of program ENGINE.

We shall soon discuss ways to counter an unbalanced moment with the addition of balance shafts, but there is a more direct approach available in this example. Figure 14-16 shows that the shaking moment is due to the action of the individual cylinders' inertial forces acting at moment arms about some center. If we consider that center to be

* Note the pattern of acceptable firing orders (FO). Write two revolutions worth of any acceptable FO, as in 1, 4, 3, 2, 1, 4, 3, 2. Any set of four successive numbers in this sequence, *either forward or backward*, is an acceptable FO. If we require the first to be cylinder 1, then the only other possibility here is the backward set 1, 2, 3, 4.

TABLE 14-3 Force and Moment Balance State of a 4-Cylinder, Inline Engine with a 0, 180, 0, 180° Crankshaft, and $z_1 = 1, z_2 = 2, z_3 = 3, z_4 = 4$

Primary forces :	$\sum_{i=1}^n \sin \phi_i = 0$	$\sum_{i=1}^n \cos \phi_i = 0$
Secondary forces :	$\sum_{i=1}^n \sin 2\phi_i = 0$	$\sum_{i=1}^n \cos 2\phi_i = 4$
Fourth harmonic forces :	$\sum_{i=1}^n \sin 4\phi_i = 0$	$\sum_{i=1}^n \cos 4\phi_i = 4$
Sixth harmonic forces :	$\sum_{i=1}^n \sin 6\phi_i = 0$	$\sum_{i=1}^n \cos 6\phi_i = 4$
Primary moments :	$\sum_{i=1}^n z_i \sin \phi_i = 0$	$\sum_{i=1}^n z_i \cos \phi_i = -2$
Secondary moments :	$\sum_{i=1}^n z_i \sin 2\phi_i = 0$	$\sum_{i=1}^n z_i \cos 2\phi_i = 6$
Fourth harmonic moments :	$\sum_{i=1}^n z_i \sin 4\phi_i = 0$	$\sum_{i=1}^n z_i \cos 4\phi_i = 6$
Sixth harmonic moments :	$\sum_{i=1}^n z_i \sin 6\phi_i = 0$	$\sum_{i=1}^n z_i \cos 6\phi_i = 6$

point *C* in the middle of the engine, it should be apparent that any primary force-balanced crankshaft design which is mirror symmetric about a transverse plane through point *C* would also have balanced primary moments as long as all cylinder spacings were uniform and all inertial forces equal. Figure 14-16a shows the 0, 180, 0, 180° crankshaft which is not mirror symmetric. The couple $F_{s1}\Delta z$ due to cylinder pairs 1, 2 has the same magnitude and sense as the couple $F_{s3}\Delta z$ due to cylinders 3 and 4, so they add. Figure 14-16b shows the 0, 180, 180, 0° crankshaft which is *mirror symmetric*. The couple $F_{s1}\Delta z$ due to cylinder pairs 1, 2 has the same magnitude but opposite sense to the couple $F_{s3}\Delta z$ due to cylinders 3, 4, so they cancel. We can then achieve both even firing and balanced primary moments by changing the sequence of crank throw phase angles to $\phi_i = 0, 180, 180, 0^\circ$ which is *mirror symmetric*.

The crank phase diagram for this design is shown in Figure 14-17. The power strokes can still be evenly spaced over 720°. A firing order of 1, 3, 4, 2 has been chosen which gives the same desired sequence of power stroke angles, $\psi_i = 0, 180, 360, 540^\circ$. (Note that a firing order of 1, 2, 4, 3 would also work with this engine.)* Equations 14.3, 14.5, and 14.7 and Table 14-4 show that the primary inertia force and primary moment are both now zero, but the secondary force, secondary moment, and inertia torque are still nonzero.

* In carbureted, inline engines a nonconsecutive firing order (i.e., not 1, 2, 3, 4) is usually preferred so that adjacent cylinders near the ends of the engine do not fire sequentially. This allows the intake manifold more time to recharge locally between intake strokes. With fuel injection, this is less critical.

To see the results of this engine configuration, run program ENGINE and select it from the *Examples* pull-down menu. Then *Plot* or *Print* the results. See Appendix A for more detailed instructions on the use of the program.

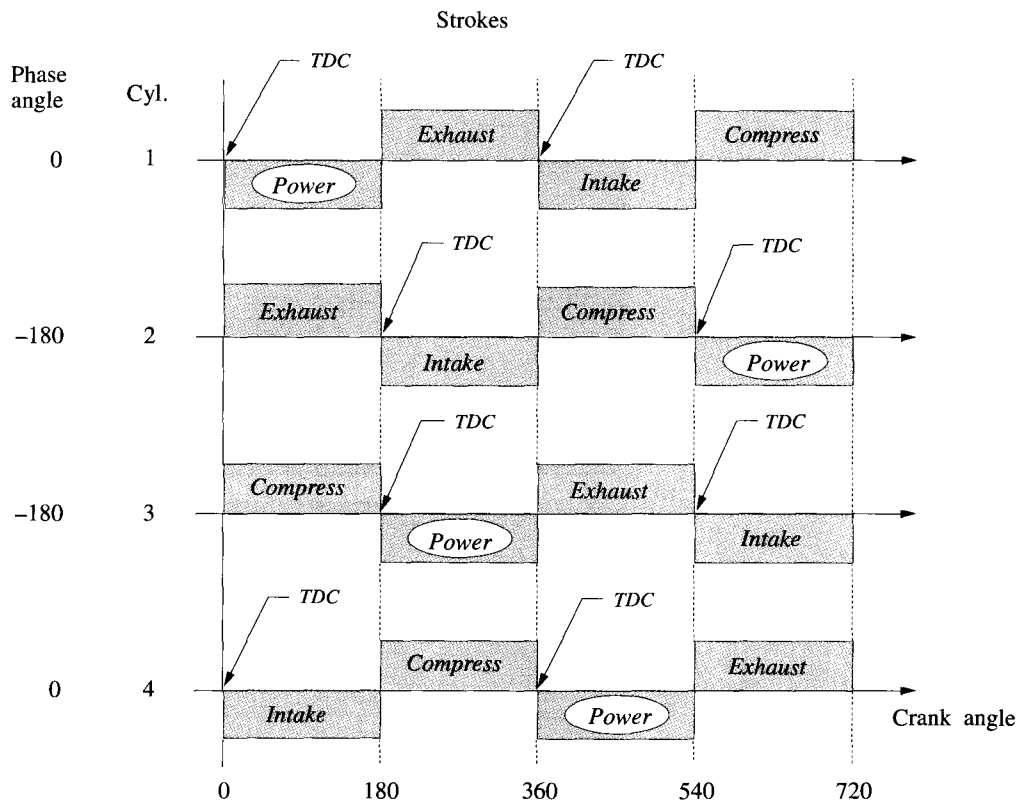


FIGURE 14-17
Even firing four-stroke, four-cylinder engine crank phase diagram with a mirror symmetric 0, 180, 180, 0° crankshaft

14.7 VEE ENGINE CONFIGURATIONS

The same design principles which apply to inline engines also apply to vee and opposed configurations. Even firing takes precedence over inertia balance and mirror symmetry of the crankshaft balances primary moments. In general, a vee engine will have similar inertia balance to that of the inline engines from which it is constructed. A vee six is essentially two three-cylinder inline engines on a common crankshaft, a vee eight is two four-cylinder inlines, etc. The larger number of cylinders allows more power pulses to be spaced out over the cycle for a smoother (and larger average) gas torque. The existence of a **vee angle** between the two inline engines introduces an additional phase shift of the inertial and gas events which is analogous to, but independent of, the phase angle effects. This vee angle is the designer’s choice, but there are good and poor choices. The same criteria of even firing and inertia balance apply to its selection.

The **vee angle** 2γ is defined as shown in Figure 14-20. Each bank is offset by its **bank angle** γ referenced to the central X axis of the engine. The crank angle ωt is measured from the X axis. Cylinder one in the right bank is the reference cylinder. Events in each bank will now be phase shifted by its bank angle as well as by the crankshaft phase

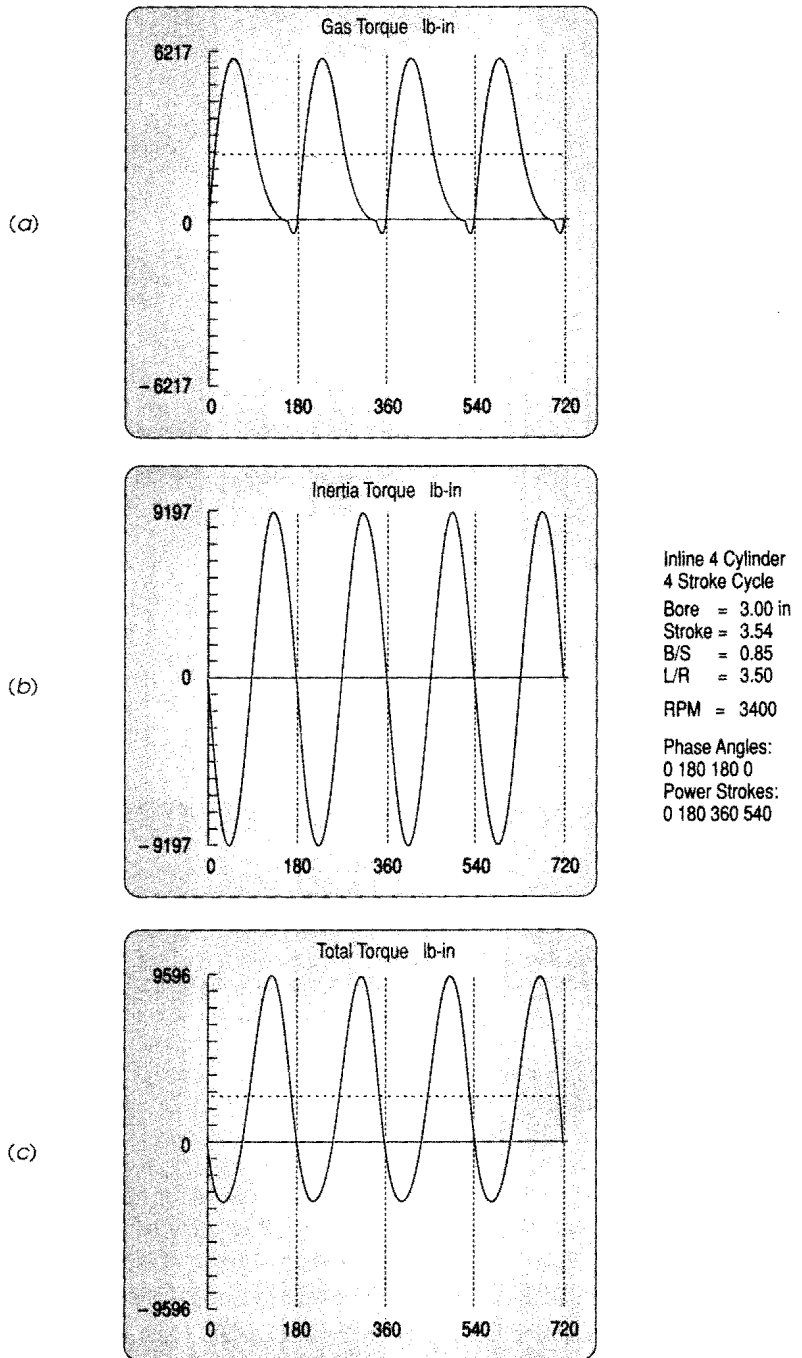


FIGURE 14-18

Torque in the four-stroke, four-cylinder inline engine with a 0, 180, 180, 0° crankshaft

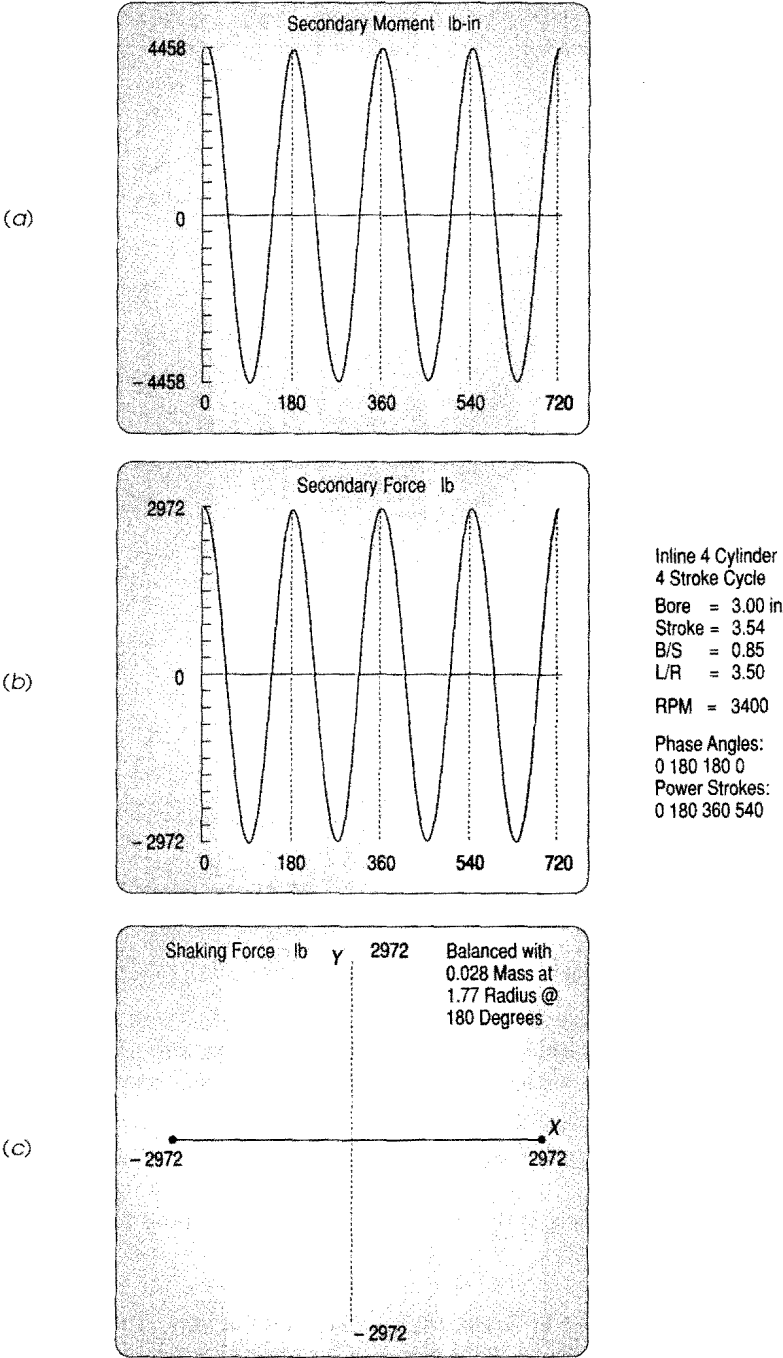


FIGURE 14-19
Shaking forces and moments in the four-stroke, four-cylinder 0, 180, 180, 0° engine

TABLE 14-4 Force and Moment Balance State of a 4-Cylinder, Inline Engine with a 0, 180, 180, 0° Crankshaft, and $z_1 = 1, z_2 = 2, z_3 = 3, z_4 = 4$

Primary forces :	$\sum_{i=1}^n \sin \phi_i = 0$	$\sum_{i=1}^n \cos \phi_i = 0$
Secondary forces :	$\sum_{i=1}^n \sin 2\phi_i = 0$	$\sum_{i=1}^n \cos 2\phi_i = 4$
Fourth harmonic forces :	$\sum_{i=1}^n \sin 4\phi_i = 0$	$\sum_{i=1}^n \cos 4\phi_i = 4$
Sixth harmonic forces :	$\sum_{i=1}^n \sin 6\phi_i = 0$	$\sum_{i=1}^n \cos 6\phi_i = 4$
Primary moments :	$\sum_{i=1}^n z_i \sin \phi_i = 0$	$\sum_{i=1}^n z_i \cos \phi_i = 0$
Secondary moments :	$\sum_{i=1}^n z_i \sin 2\phi_i = 0$	$\sum_{i=1}^n z_i \cos 2\phi_i = 6$
Fourth harmonic moments :	$\sum_{i=1}^n z_i \sin 4\phi_i = 0$	$\sum_{i=1}^n z_i \cos 4\phi_i = 6$
Sixth harmonic moments :	$\sum_{i=1}^n z_i \sin 6\phi_i = 0$	$\sum_{i=1}^n z_i \cos 6\phi_i = 6$

angles. These two phase shifts will superpose. Taking any one cylinder in either bank as an example, let its instantaneous crank angle be represented by:

$$\theta = (\omega t - \phi_i) \quad (14.10a)$$

Consider first a two-cylinder vee engine with one cylinder in each bank and with both sharing a common crank throw. The shaking force for a single cylinder in the direction of piston motion $\hat{\mathbf{u}}$ with θ measured from the piston axis is:

$$\mathbf{F}_s \cong m_B r \omega^2 \left(\cos \theta + \frac{r}{l} \cos 2\theta \right) \hat{\mathbf{u}} \quad (14.10b)$$

The total shaking force is the vector sum of the contributions from each bank.

$$\mathbf{F}_s = \mathbf{F}_{s_L} + \mathbf{F}_{s_R} \quad (14.10c)$$

We now want to refer the crank angle to the central X axis. The shaking forces for the right (R) and left (L) banks, in the planes of the respective cylinder banks can then be expressed as:

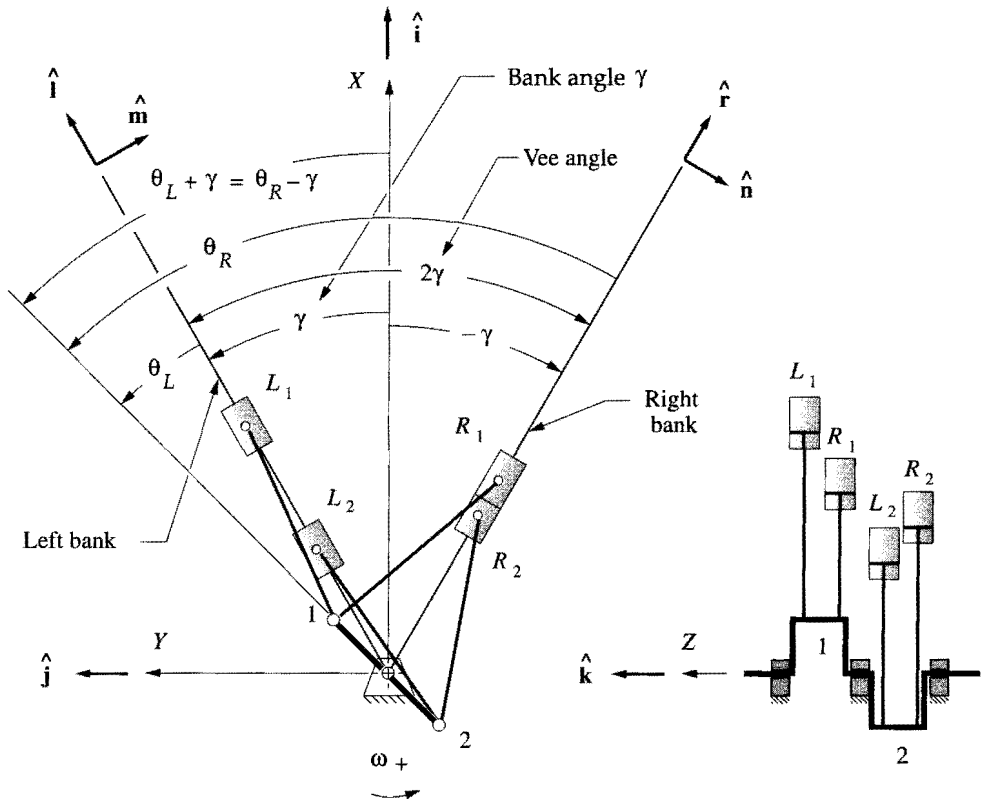


FIGURE 14-20

Vee-engine geometry

$$\mathbf{F}_{sR} \cong m_B r \omega^2 \left[\cos(\theta + \gamma) + \frac{r}{l} \cos 2(\theta + \gamma) \right] \hat{\mathbf{r}} \quad (14.10d)$$

$$\mathbf{F}_{sL} \cong m_B r \omega^2 \left[\cos(\theta - \gamma) + \frac{r}{l} \cos 2(\theta - \gamma) \right] \hat{\mathbf{l}}$$

Note that the bank angle γ is added to or subtracted from the crank angle for each cylinder bank to reference it to the central X axis. The forces are still directed along the planes of the cylinder banks. Substitute the identities:

$$\begin{aligned} \cos(\theta + \gamma) &= \cos\theta \cos\gamma - \sin\theta \sin\gamma \\ \cos(\theta - \gamma) &= \cos\theta \cos\gamma + \sin\theta \sin\gamma \end{aligned} \quad (14.10e)$$

to get:

$$\begin{aligned}
 \mathbf{F}_{s_R} &\cong m_B r \omega^2 \left[\begin{array}{c} \cos \theta \cos \gamma - \sin \theta \sin \gamma \\ + \frac{r}{l} (\cos 2\theta \cos 2\gamma - \sin 2\theta \sin 2\gamma) \end{array} \right] \hat{\mathbf{r}} \\
 \mathbf{F}_{s_L} &\cong m_B r \omega^2 \left[\begin{array}{c} \cos \theta \cos \gamma + \sin \theta \sin \gamma \\ + \frac{r}{l} (\cos 2\theta \cos 2\gamma + \sin 2\theta \sin 2\gamma) \end{array} \right] \hat{\mathbf{i}}
 \end{aligned} \quad (14.10f)$$

Now, to account for the possibility of multiple cylinders, phase shifted within each bank, substitute equation 14.10a for θ and replace the sums of angle terms with products from the identities:

$$\begin{aligned}
 \cos(\omega t - \phi_i) &= (\cos \omega t \cos \phi_i + \sin \omega t \sin \phi_i) \\
 \sin(\omega t - \phi_i) &= (\sin \omega t \cos \phi_i - \cos \omega t \sin \phi_i)
 \end{aligned} \quad (14.10g)$$

After much manipulation, the expressions for the contributions from right and left banks reduce to:

for the right bank:

$$\mathbf{F}_{s_R} \cong m_B r \omega^2 \left[\begin{array}{c} (\cos \omega t \cos \gamma - \sin \omega t \sin \gamma) \sum_{i=1}^{n/2} \cos \phi_i \\ + (\cos \omega t \sin \gamma + \sin \omega t \cos \gamma) \sum_{i=1}^{n/2} \sin \phi_i \\ + \frac{r}{l} (\cos 2\omega t \cos 2\gamma - \sin 2\omega t \sin 2\gamma) \sum_{i=1}^{n/2} \cos 2\phi_i \\ + \frac{r}{l} (\cos 2\omega t \sin 2\gamma + \sin 2\omega t \cos 2\gamma) \sum_{i=1}^{n/2} \sin 2\phi_i \end{array} \right] \hat{\mathbf{r}} \quad (14.10h)$$

and for the left bank:

$$\mathbf{F}_{s_L} \cong m_B r \omega^2 \left[\begin{array}{c} (\cos \omega t \cos \gamma + \sin \omega t \sin \gamma) \sum_{i=n/2+1}^n \cos \phi_i \\ - (\cos \omega t \sin \gamma - \sin \omega t \cos \gamma) \sum_{i=n/2+1}^n \sin \phi_i \\ + \frac{r}{l} (\cos 2\omega t \cos 2\gamma + \sin 2\omega t \sin 2\gamma) \sum_{i=n/2+1}^n \cos 2\phi_i \\ - \frac{r}{l} (\cos 2\omega t \sin 2\gamma - \sin 2\omega t \cos 2\gamma) \sum_{i=n/2+1}^n \sin 2\phi_i \end{array} \right] \hat{\mathbf{i}} \quad (14.10i)$$

The summations in equations 14.10h and i give a set of sufficient criteria for **zero shaking force** through the second harmonic for each bank, similar to those for the inline engine in equation 14.3 (p. 648). We can resolve the shaking forces for each bank into components along and normal to the central X axis of the vee engine:

$$\begin{aligned} F_{s_x} &= (F_{s_L} + F_{s_R}) \cos \gamma \hat{\mathbf{i}} \\ F_{s_y} &= (F_{s_L} - F_{s_R}) \sin \gamma \hat{\mathbf{j}} \\ \mathbf{F}_s &= F_{s_x} \hat{\mathbf{i}} + F_{s_y} \hat{\mathbf{j}} \end{aligned} \quad (14.10j)$$

Equation 14.10j provides additional opportunities for cancellation of shaking forces beyond the choice of phase angles; e.g., even with nonzero values of F_{s_L} and F_{s_R} , if γ is 90° , then the x component of the shaking force will be zero. Also, if $F_{s_L} = F_{s_R}$, then the y component of the shaking force will be zero for any γ . This situation obtains for the case of a horizontally opposed engine (see Section 14.8, p. 674). With some vee or opposed engines it is possible to get cancellation of shaking force components even when the summations in equation 14.10 are not all zero.

The **shaking moment** equations are easily formed from the shaking force equations by multiplying each term in the summations by the moment arm as was done in equations 14.6 (p. 650). The moments exist within each bank and their vectors will be orthogonal to the respective cylinder planes. For the right bank we define a moment unit vector $\hat{\mathbf{n}}$ perpendicular to the $\mathbf{r}$ *hat*- Z plane in Figure 14-20 (p. 665). For the left bank we define a moment unit vector $\hat{\mathbf{m}}$ perpendicular to the $\mathbf{l}$ *hat*- Z plane in Figure 14-20.

$$\mathbf{M}_{s_R} \cong m_B r \omega^2 \begin{bmatrix} (\cos \omega t \cos \gamma - \sin \omega t \sin \gamma) \sum_{i=1}^{n/2} z_i \cos \phi_i \\ + (\cos \omega t \sin \gamma + \sin \omega t \cos \gamma) \sum_{i=1}^{n/2} z_i \sin \phi_i \\ + \frac{r}{l} (\cos 2\omega t \cos 2\gamma - \sin 2\omega t \sin 2\gamma) \sum_{i=1}^{n/2} z_i \cos 2\phi_i \\ + \frac{r}{l} (\cos 2\omega t \sin 2\gamma + \sin 2\omega t \cos 2\gamma) \sum_{i=1}^{n/2} z_i \sin 2\phi_i \end{bmatrix} \hat{\mathbf{n}} \quad (14.11a)$$

$$\mathbf{M}_{s_L} \cong m_B r \omega^2 \begin{bmatrix} (\cos \omega t \cos \gamma + \sin \omega t \sin \gamma) \sum_{i=n/2+1}^n z_i \cos \phi_i \\ - (\cos \omega t \sin \gamma - \sin \omega t \cos \gamma) \sum_{i=n/2+1}^n z_i \sin \phi_i \\ + \frac{r}{l} (\cos 2\omega t \cos 2\gamma + \sin 2\omega t \sin 2\gamma) \sum_{i=n/2+1}^n z_i \cos 2\phi_i \\ - \frac{r}{l} (\cos 2\omega t \sin 2\gamma - \sin 2\omega t \cos 2\gamma) \sum_{i=n/2+1}^n z_i \sin 2\phi_i \end{bmatrix} \hat{\mathbf{m}} \quad (14.11b)$$

The summations provide a set of sufficient criteria for **zero shaking moment** through the second harmonic for each bank, similar to those found for the inline engine in equation 14.7. Resolving the shaking moments for each bank into components along and normal to the central X axis of the vee engine gives:

$$\begin{aligned} M_{s_x} &= (M_{s_L} - M_{s_R}) \sin \gamma \\ M_{s_y} &= (-M_{s_L} - M_{s_R}) \cos \gamma \\ \mathbf{M}_s &= M_{s_x} \hat{\mathbf{i}} + M_{s_y} \hat{\mathbf{j}} \end{aligned} \quad (14.11c)$$

Equation 14.11c allows possible cancellation of shaking moment components for some vee or opposed configurations even when the summations in equation 14.11 are not all zero; e.g., if γ is 90° , then the y component of the shaking moment is zero.

The **inertia torques** from the right and left banks of a vee engine are:

$$\mathbf{T}_{i21_R} \cong \frac{1}{2} m_B r^2 \omega^2 \left[\begin{aligned} &\frac{r}{2l} \left(\sin(\omega t + \gamma) \sum_{i=1}^{n/2} \cos \phi_i - \cos(\omega t + \gamma) \sum_{i=1}^{n/2} \sin \phi_i \right) \\ &- \left(\sin 2(\omega t + \gamma) \sum_{i=1}^{n/2} \cos 2\phi_i - \cos 2(\omega t + \gamma) \sum_{i=1}^{n/2} \sin 2\phi_i \right) \\ &- \frac{3r}{2l} \left(\sin 3(\omega t + \gamma) \sum_{i=1}^{n/2} \cos 3\phi_i - \cos 3(\omega t + \gamma) \sum_{i=1}^{n/2} \sin 3\phi_i \right) \end{aligned} \right] \hat{\mathbf{k}} \quad (14.12a)$$

$$\mathbf{T}_{i21_L} \cong \frac{1}{2} m_B r^2 \omega^2 \left[\begin{aligned} &\frac{r}{2l} \left(\sin(\omega t - \gamma) \sum_{i=n/2+1}^n \cos \phi_i - \cos(\omega t - \gamma) \sum_{i=n/2+1}^n \sin \phi_i \right) \\ &- \left(\sin 2(\omega t - \gamma) \sum_{i=n/2+1}^n \cos 2\phi_i - \cos 2(\omega t - \gamma) \sum_{i=n/2+1}^n \sin 2\phi_i \right) \\ &- \frac{3r}{2l} \left(\sin 3(\omega t - \gamma) \sum_{i=n/2+1}^n \cos 3\phi_i - \cos 3(\omega t - \gamma) \sum_{i=n/2+1}^n \sin 3\phi_i \right) \end{aligned} \right] \hat{\mathbf{k}} \quad (14.12b)$$

Add the contributions from each bank for the total. For **zero inertia torque** through the third harmonic in a vee engine it is sufficient (but not necessary) that:

$$\begin{array}{cccc}
\sum_{i=1}^{n/2} \sin \phi_i = 0 & \sum_{i=1}^{n/2} \cos \phi_i = 0 & \sum_{i=n/2+1}^n \sin \phi_i = 0 & \sum_{i=n/2+1}^n \cos \phi_i = 0 \\
\sum_{i=1}^{n/2} \sin 2\phi_i = 0 & \sum_{i=1}^{n/2} \cos 2\phi_i = 0 & \sum_{i=n/2+1}^n \sin 2\phi_i = 0 & \sum_{i=n/2+1}^n \cos 2\phi_i = 0 \\
\sum_{i=1}^{n/2} \sin 3\phi_i = 0 & \sum_{i=1}^{n/2} \cos 3\phi_i = 0 & \sum_{i=n/2+1}^n \sin 3\phi_i = 0 & \sum_{i=n/2+1}^n \cos 3\phi_i = 0
\end{array} \quad (14.12c)$$

Note that when equations 14.12a and b are added, particular combinations of ϕ_i and γ may cancel the inertia torque even when some terms of equation 14.12c are nonzero.

The gas torque is:

$$T_{g21} \cong F_g r \sum_{i=1}^n \left(\sin[\omega t - (\psi_i + \gamma_k)] \left\{ 1 + \frac{r}{l} \cos[\omega t - (\psi_i + \gamma_k)] \right\} \right) \hat{\mathbf{k}} \quad (14.13)$$

where the left bank has a *bank angle*, $\gamma_k = +\gamma$ and the right bank a *bank angle*, $\gamma_k = -\gamma$.

It is possible to design a vee engine which has as many crank throws as cylinders, but, for several reasons, this is not always done. The principal advantage of a vee engine over an inline of the same number of cylinders is its more compact size and greater stiffness. It can be about half the length of a comparable inline engine (at the expense of greater width), provided that the crankshaft is designed to accommodate two conrods per crank throw as shown in Figure 14-21. Cylinders in opposite banks then share a crank throw. One bank of cylinders is shifted along the crankshaft axis by the thickness of a conrod. The shorter, wider cylinder block and the shorter crankshaft are much stiffer in both torsion and bending than are those for an inline engine with the same number of cylinders. Figure 14-22 shows computer simulations of several bending and one torsional mode of vibration for a four throw crankshaft. The deflections are greatly exaggerated. The necessarily contorted shape of a crankshaft makes it difficult to control these deflections by design. If excessive in magnitude, they can lead to structural failure.

As an example we will now design the crankshaft for a four-stroke cycle, vee-eight engine. We could put two $\phi_i = 0, 180, 180, 0^\circ$ four-cylinder engines together on one such crankshaft and have the same balance conditions as the four-stroke, four-cylinder engine designed in the previous section (primaries balanced, secondaries unbalanced). However, the motivation for choosing that crankshaft for the four-cylinder engine was the need to space the four available power pulses evenly across the cycle. Equation 14.8b (p. 652) then dictated a 180° delta phase angle for that engine. Now we have eight cylinders available and equation 14.8b defines a delta phase angle of 90° for optimum power pulse spacing. This means we could use the $\phi_i = 0, 90, 180, 270^\circ$ crankshaft designed for the two-stroke four-cylinder engine shown in Figure 14-11 (p. 654), and take advantage of its better inertia balance condition as well as achieve even firing in the four-stroke eight-cylinder vee engine.

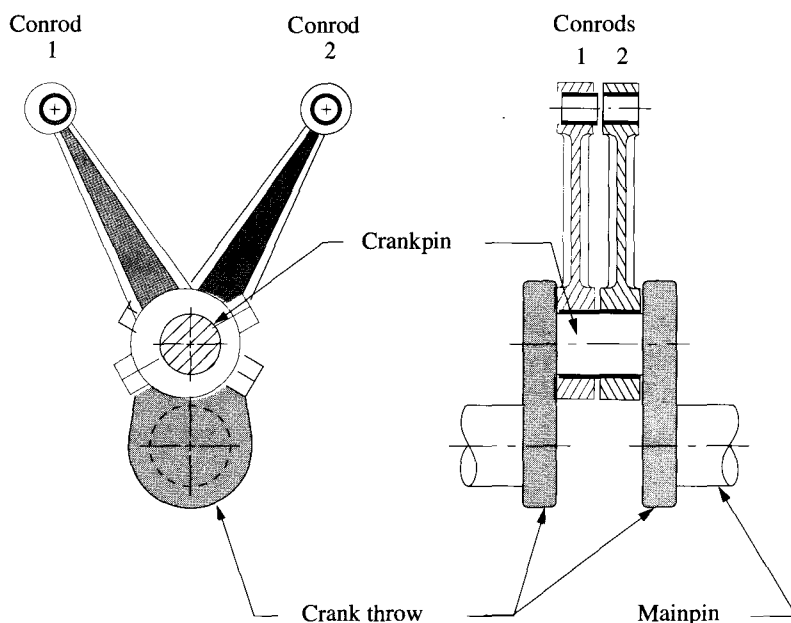


FIGURE 14-21

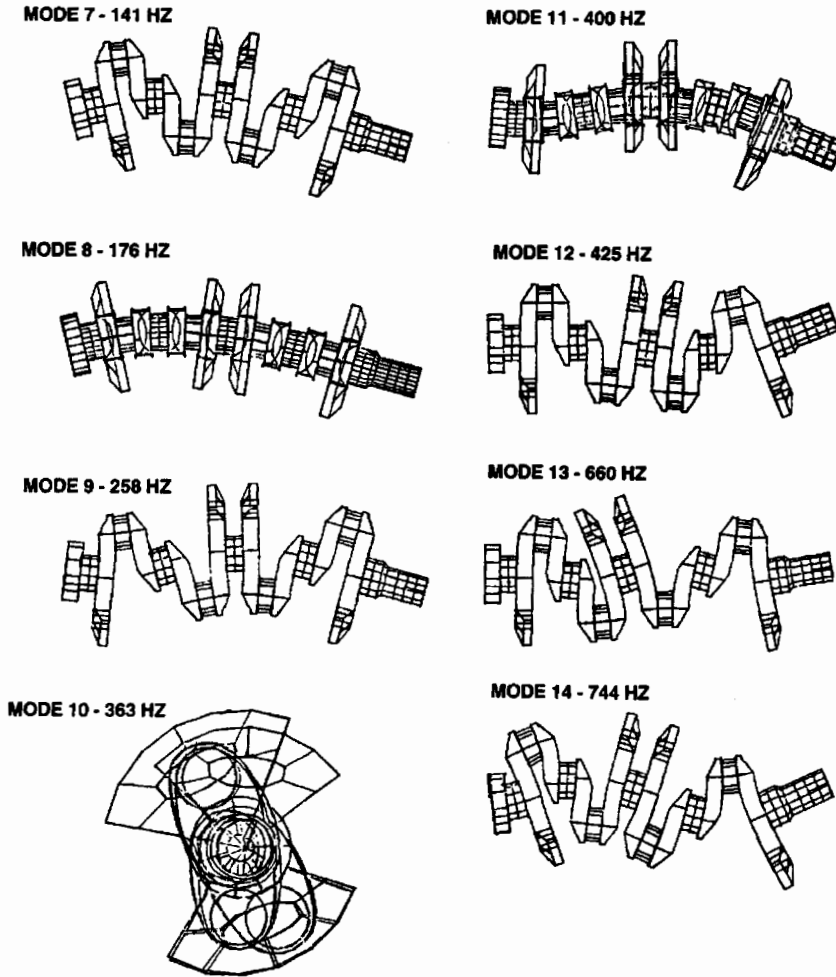
Two connecting rods on a common crank throw

* The explanation for this is quite simple. Equation 14.7b (p. 650) shows that second moments are a function of twice the phase angles and the cylinder moment arms. If you double the values of the original 0, 90, 180, 270° phase angle sequence and modulo them with 360, you get 0, 180, 0, 180° which is not mirror symmetric. Doubling the new phase angle sequence of 0, 90, 270, 180°, modulo 360 gives 0, 180, 180, 0° which is mirror symmetric. It is this symmetry of the doubled phase angles that causes cancellation of the second harmonic of the shaking moment.

The $\phi_i = 0, 90, 180, 270^\circ$ four-cylinder crankshaft has all inertia factors equal to zero except for the primary and secondary moments. We learned that arranging the crank throws with mirror symmetry about the midplane would balance the primary moment. Some thought and/or sketches will reveal that it is not possible to obtain this mirror symmetry with any four throw, 90° delta phase angle crankshaft arrangement. However, just as rearranging the order of the crank throws from $\phi_i = 0, 180, 0, 180^\circ$ to $\phi_i = 0, 180, 180, 0^\circ$ had an effect on the shaking moments, rearranging this crankshaft's throw-order will as well. A crankshaft of $\phi_i = 0, 90, 270, 180^\circ$ has all inertia factors equal to zero except for the primary moment. The secondary moment is now gone.* This is an advantage worth taking. We will use this crankshaft for the vee eight and deal with the primary moment later.

Figure 14-23a shows the crank phase diagram for the right bank of a vee-eight engine with a $\phi_i = 0, 90, 270, 180^\circ$ crankshaft. Figure 14-23b shows the crank phase diagram for the second (left) bank which is identical to that of the right bank (as it must be since they share crank throws), but it is *shifted to the right by the vee angle 2γ* . Note that in Figure 14-20 (p. 665), the two pistons are driven by conrods on a common crank throw with positive ω , and the piston in the right bank will reach TDC before the one in the left bank. Thus as we show it, the left bank's piston motions lag those of the right bank. Lagging events occur later in time, so we must shift the second (left) bank rightward by the vee angle on the crank phase diagram.

We would like to shift the second bank of cylinders such that its power pulses are evenly spaced among those of the first bank. A little thought (and reference to equation

**FIGURE 14-22**

Bending and torsional modes of vibration in a four-throw crankshaft
 Courtesy of Chevrolet Division, General Motors Corp.

14.8b, p. 652) should reveal that, in this example, each four-cylinder bank has potentially $720/4 = 180^\circ$ between power pulses. Our chosen crank throws are spaced at 90° increments. A 90° bank angle will be optimum in this case as the phase angles and bank angles will add to create an effective spacing of 180° . Every vee engine design will have one or more optimum vee angles that will give approximately even firing with any particular set of crank phase angles.

Several firing orders are possible with this many cylinders. Vee engines are often arranged to fire cylinders in opposite banks successively to balance the fluid flow demands in the intake manifold. Our cylinders are numbered from front to back, first down the right bank and then down the left bank. The firing order shown in Figure 14-23b is

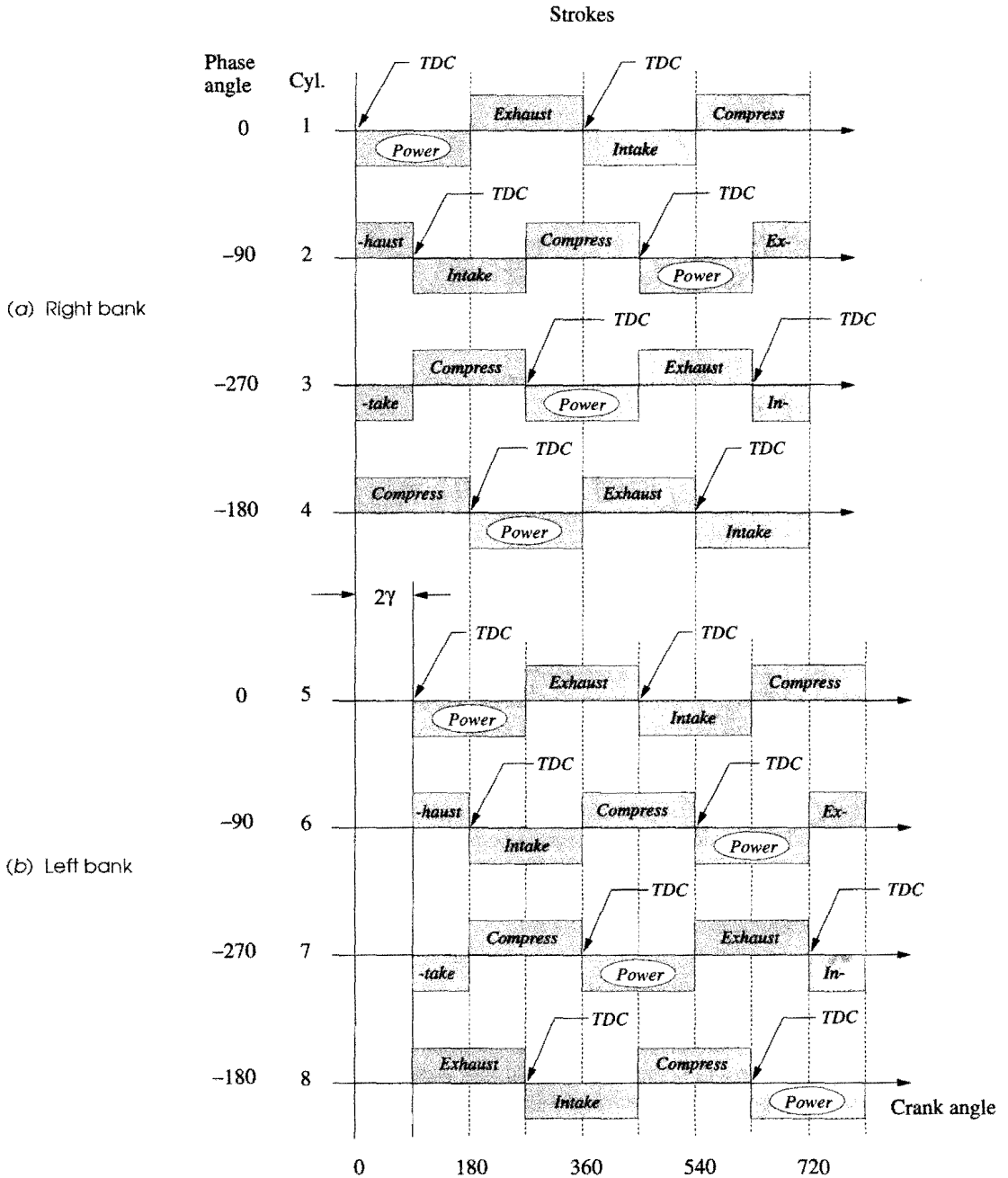


FIGURE 14-23

Four-stroke vee-eight crank phase diagram with 0, 90, 270, 180° crankshaft phase angles

90 deg Vee 8 Cylinder
4 Stroke Cycle
Bore = 2.50 in
Stroke = 2.55
B/S = 0.98
L/R = 3.50
RPM = 3400
Phase Angles:
0 90 270 180 0 90 270 180
Power Strokes:
0 90 180 270 360 450 540 630

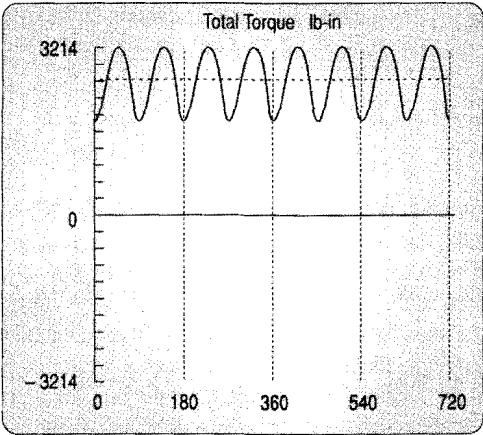


FIGURE 14-24

Total torque in the 90° V-8 engine with 0, 90, 270, 180° crankshaft phase angles

1, 5, 4, 3, 7, 2, 6, 8 and results in power stroke angles $\psi_i = 0, 90, 180, 270, 360, 450, 540, 630^\circ$. This will clearly give even firing with a power pulse every 90° .

Figure 14-24 shows the total torque for this engine design. The total torque in this case is equal to the gas torque as the inertia torque is zero. Table 14-5 and Figure 14-25 show the only significant unbalanced inertial component in this engine to be the primary moment, which is quite large. The fourth harmonic terms have negligible coefficients in the Fourier series, and we have truncated them from the equations. We will address the balancing of this primary moment in a later section of this chapter.

90 deg Vee 8 Cylinder
4 Stroke Cycle
Bore = 2.50 in
Stroke = 2.55
B/S = 0.98
L/R = 3.50
RPM = 3400
Phase Angles:
0 90 270 180 0 90 270 180
Power Strokes:
0 90 180 270 360 450 540 630

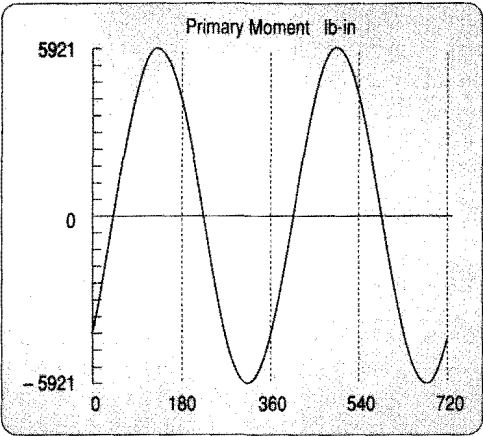


FIGURE 14-25

Unbalanced primary moment in the 90° V-8 engine with a 0, 90, 270, 180° crankshaft

TABLE 14-5 Force and Moment Balance State of an 8-Cylinder, Vee Engine with a 0, 90, 270, 180° Crankshaft, and $z_1 = 1, z_2 = 2, z_3 = 3, z_4 = 4$

Primary forces :	$\sum_{i=1}^n \sin \phi_i = 0$	$\sum_{i=1}^n \cos \phi_i = 0$
Secondary forces :	$\sum_{i=1}^n \sin 2\phi_i = 0$	$\sum_{i=1}^n \cos 2\phi_i = 0$
Fourth harmonic forces :	$\sum_{i=1}^n \sin 4\phi_i = 0$	$\sum_{i=1}^n \cos 4\phi_i = 8$
Sixth harmonic forces :	$\sum_{i=1}^n \sin 6\phi_i = 0$	$\sum_{i=1}^n \cos 6\phi_i = 0$
Primary moments :	$\sum_{i=1}^n z_i \sin \phi_i = -4$	$\sum_{i=1}^n z_i \cos \phi_i = -2$
Secondary moments :	$\sum_{i=1}^n z_i \sin 2\phi_i = 0$	$\sum_{i=1}^n z_i \cos 2\phi_i = 0$
Fourth harmonic moments :	$\sum_{i=1}^n z_i \sin 4\phi_i = 0$	$\sum_{i=1}^n z_i \cos 4\phi_i = 28$
Sixth harmonic moments :	$\sum_{i=1}^n z_i \sin 6\phi_i = 0$	$\sum_{i=1}^n z_i \cos 6\phi_i = 0$

Any vee cylinder configuration may have one or more desirable vee angles which will give both even firing and acceptable inertia balance. However, vee engines of fewer than twelve cylinders will not be completely balanced by means of their crankshaft configuration. The desirable vee angles will typically be an integer multiple (including one) or submultiple of the optimum delta phase angle as defined in equations 14.8 (pp. 652 and 654) for that engine. Ninety degrees is the optimum vee angle (2γ) for an eight-cylinder vee engine. To see the results for this vee-eight engine configuration, run program ENGINE and select the vee eight from the *Examples* pull-down menu. See Appendix A for instructions on the use of the program.

14.8 OPPOSED ENGINE CONFIGURATIONS

An opposed engine is essentially a vee engine with a 180° vee angle. The advantage, particularly with a small number of cylinders such as two or four, is the relatively good balance condition possible. A four-stroke opposed twin* with 0, 180° crank has even firing plus primary force balance, though the primary moment and all higher harmonics of force and moment are nonzero. A four-stroke, opposed four-cylinder engine (flat four) with a 0, 180, 180, 0°, four-throw crank has primary force balance but must fire its cyl-

* As in the BMW motorcycle.

inders in pairs, so its firing pattern looks like a twin. A four-stroke flat four with a two-throw, 0, 180° crank will have even firing and the same balance condition as the inline four with a 0, 180, 180, 0° crank. Program ENGINE will calculate the parameters for opposed as well as vee and inline configurations.

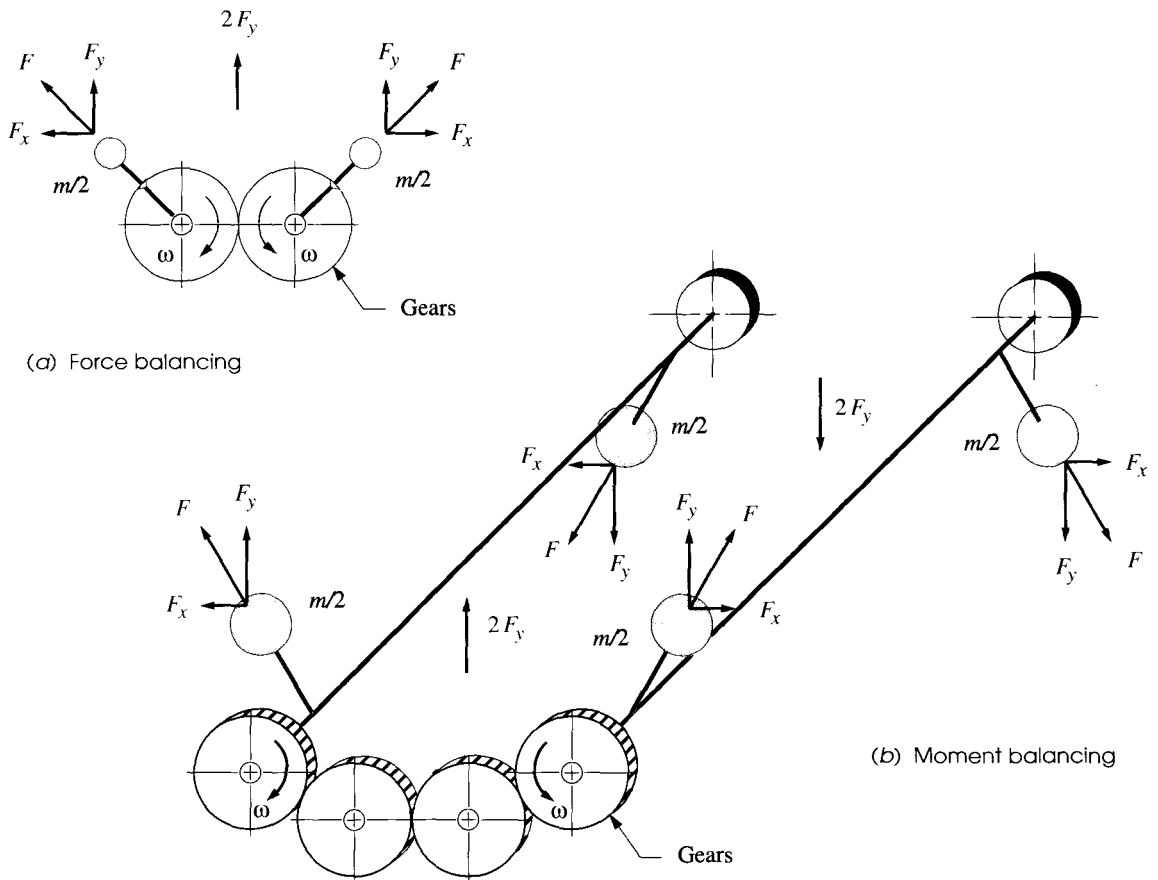
14.9 BALANCING MULTICYLINDER ENGINES

With a sufficient number (n) of cylinders, properly arranged, an engine can be inherently balanced. In a two-stroke engine with its crank throws arranged for even firing, all harmonics of shaking force will be balanced except those whose harmonic number is a multiple of n . In a four-stroke engine with its crank throws arranged for even firing, all harmonics of shaking force will be balanced except those whose harmonic number is a multiple of $n/2$. Primary shaking moment components will be balanced if the crankshaft is mirror symmetric about the central transverse plane. A four-stroke inline configuration then requires at least six cylinders to be inherently balanced through the fourth harmonic. We have seen that an inline four with a 0, 180, 180, 0° crankshaft has nonzero secondary forces and moments as well as nonzero inertia torque. The inline six with a mirror symmetric crank of $\phi_i = 0, 240, 120, 120, 240, 0^\circ$ will have zero shaking forces and moments through the fourth harmonic, though the inertia torque's third harmonic will still be present. To see the results of this six-cylinder inline engine configuration, run program ENGINE and select the inline six from the *Examples* pull-down menu.

A vee twelve is then the smallest vee engine with this inherent state of near perfect balance, as it is two inline sixes on a common crankshaft. We have seen that vee engines take on the balance characteristics of the inline banks from which they are made. Equations 14.10 and 14.11 (pp. 664 to 668) introduced no new criteria for balance in the vee engine over those already defined in equations 14.3 (p. 648) and 14.5 (p. 650) for shaking force and moment balance in the inline engine. Open the file BMWV12.ENG in program ENGINE to see the results for a vee-twelve engine. The common vee-eight engine with crankshaft phase angles of $\phi_i = 0, 90, 270, 180^\circ$ has an unbalanced primary moment as does the inline four from which it is made. It is an example in program ENGINE.

A vee-six engine with 0, 240, 120° crankshaft has unbalanced primary and secondary moments as does the three-cylinder inline from which it is made. This vee six needs a 120° vee angle for proper balance. To reduce engine width, vee sixes are most often made with a 60° vee angle which gives even firing with a 6-throw crank. Some use 90° vee angles to allow their assembly on the same production-line tooling as 90° vee eights, but 90° vee sixes will run rough due to uneven firing unless the crankshaft is redesigned to shift (or splay) the two conrods on each pin by 30°. This results in a more complicated 6-throw crankshaft but gives an even firing, but dynamically unbalanced engine.

Unbalanced inertia torques can be smoothed with a flywheel as was shown in Section 13.10 (p. 631) for the single-cylinder engine. Note that even an engine having zero inertial torque may require a flywheel to smooth its variations in gas torque. The total torque function should be used to determine the energy variations to be absorbed by a flywheel as it contains both gas torque and inertia torque (if any). The method of Section 11.11 (p. 548) also applies to calculation of the flywheel size needed in an engine, based on its variation in the total torque function. Program ENGINE will compute the areas under the total torque pulses needed for the calculation. See the referenced sections for the proper flywheel design procedure.

**FIGURE 14-26**

Counterrotating eccentric masses can balance forces and moments

* This is called a Lanchester balancer after its English inventor who developed it for aircraft engines prior to World War I (c. 1913). It is still used in various kinds of machinery as well as in engines to cancel inertia forces.

Unbalanced shaking forces and shaking moments can be cancelled by the addition of one or more rotating balance shafts within the engine. To cancel the primary components requires two balance shafts rotating at crank speed, one of which can be the crankshaft itself. To cancel the secondary components usually requires at least two balance shafts rotating at twice crank speed, gear or chain driven from the crankshaft. Figure 14-26a shows a pair of counterrotating shafts which are fitted with eccentric masses arranged 180° out of phase.* As shown, the unbalanced centrifugal forces from the equal, unbalanced masses will add to give a shaking force component in the vertical direction of twice the unbalanced force from each mass, while their horizontal components will exactly cancel. Pairs of counterrotating eccentrics can be arranged to provide a harmonically varying force in anyone plane. The harmonic frequency will be determined by the rotational speed of the shafts.

If we arrange two pairs of eccentrics, with one pair displaced some distance along the shaft from the other, and also rotated 180° around the shaft from the first, as shown

in Figure 14-26b, we will get a harmonically varying couple in one plane. There will be cancellation of the forces in one direction and summation in an orthogonal direction.

Thus to cancel the shaking moment in any plane, we can arrange a pair of shafts, each containing two eccentric masses displaced along those shafts, 180° out of phase, and gear the shafts together to rotate in opposite directions at any multiple of crankshaft speed. To cancel the shaking force as well, it is only necessary to provide sufficient additional unbalanced mass in one of the pairs of eccentric masses to generate a shaking force opposite to that of the engine, over and above that needed to generate the forces of the couple.

In an inline engine, the unbalanced forces and moments are all confined to the single plane of the cylinders as they are due entirely to the reciprocating masses assumed concentrated at the wrist pin. (We are assuming that all crank throws are exactly balanced rotationally to cancel the effects of mass at the crankpin.) In a vee engine, however, the shaking forces and moments have both x and y components as shown in equations 14.10 and 14.11 (pp. 664 to 668) and in Figure 14-20 (p. 665). Each bank's piston's shaking effects are acting within the plane of that bank's cylinders, and the bank angle γ is used to resolve them into x and y components. Figure 14-27 shows the two-dimensional shaking force present in a two-cylinder 90° vee engine. The two conrods share a common crank throw at a phase angle of zero in this particular engine. The force due to each piston is confined to the reciprocating plane (bank) of that piston, but the vee angle between the cylinder banks creates the pattern shown when the primary and secondary components of each piston force are added together in their proper phase relationship.

The 90° vee eight engine with a 0, 90, 270, 180° crankshaft, which has only an unbalanced primary moment, presents a special case. The 90° angle between the banks results in equal horizontal and vertical components of the primary shaking moment which reduces it to a couple of constant magnitude rotating about the crank axis at crankshaft speed as shown in Figure 14-28, which is a view looking at the end of the crankshaft axis.

90 deg Vee 2 Cylinder
4 Stroke Cycle
Bore = 3.00 in
Stroke = 3.54
B/S = 0.85
L/R = 3.50
RPM = 3400

Phase Angles:
0 0
Power Strokes:
0 270

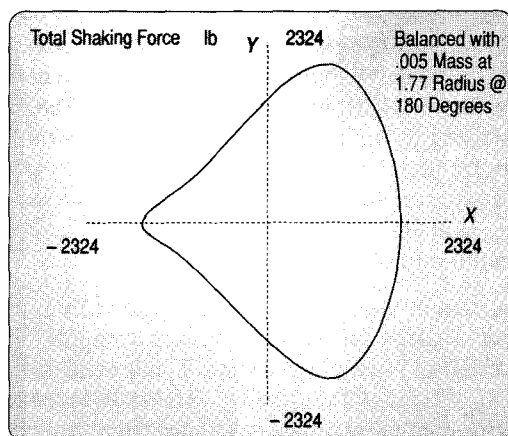


FIGURE 14-27

Shaking force in a 90° vee twin engine (looking end-on to the crankshaft axis)

90 deg Vee 8 Cylinder
 4 Stroke Cycle
 Bore = 2.50 in
 Stroke = 2.55
 B/S = 0.98
 L/R = 3.50
 RPM = 3400
 Phase Angles:
 0 90 270 180 0 90 270 180
 Power Strokes:
 0 90 180 270 360 450 540 630

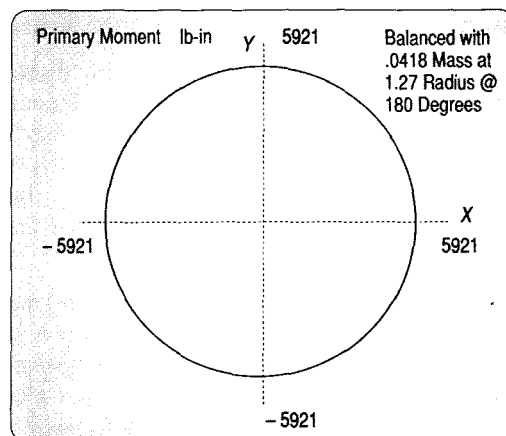


FIGURE 14-28

Primary moment in the 90° vee-eight engine (looking end-on to the crankshaft axis)

With this vee-eight engine, the primary moment can be balanced by merely adding two eccentric counterweights of proper size and opposite orientation to the crankshaft alone. No independent, second balance shaft is needed in the 90° vee-eight engine with this crankshaft. The 180°-out-of-phase counterweights are typically placed near the ends of the crankshaft to obtain the largest moment arm possible and thus reduce their size.

The shaking force of the 90° vee twin depicted in Figure 14-27 (p. 677) has a rotating primary component of constant magnitude that also can be cancelled with counterweights on the crankshaft alone. However, its second harmonic is planar (in the YZ plane). To cancel it requires a pair of twice-speed balance shafts as shown in Figure 14-26a (p. 676).

The 60° vee-six engine with a 0, 240, 120, 60, 300, 180° crankshaft has unbalanced primary and secondary moments, each of which is a constant magnitude rotating vector like that of the vee eight shown in Figure 14-28. The primary component can be completely balanced by adding counterweights to the crankshaft as in the 90° vee eight. Some manufacturers* also add a single balance shaft in the valley of the vee six, driven by gears at twice crankshaft speed to cancel the constant magnitude second harmonic of the shaking moment. This combination results in complete cancellation of the shaking moment in this engine.

Calculation of the magnitude and location of the eccentric balance masses needed to cancel any shaking forces or moments is a straightforward exercise in **static balancing** (for forces) and **two-plane dynamic balancing** (for moments) as discussed in Sections 12.1 (p. 571) and 12.2 (p. 574). The unbalanced forces and moments for the particular engine configuration are calculated from the appropriate equations in this chapter. Two correction planes must be selected along the length of the balance shafts/crankshaft being designed. The magnitude and angular locations of the balance masses can then be calculated by the methods described in the noted sections of Chapter 12.

* General Motors Corp.,
 Buick V6.

Secondary Balance in the Four Cylinder Inline Engine

The four-cylinder inline engine with a 0, 180, 180, 0° crankshaft is one of the most widely used engines in the automobile industry. As described in a previous section, this engine suffers from unbalanced secondary force, moment and torque. If the displacement of the engine is less than about 2.0 liters, then the magnitudes of the secondary forces may be small enough to be ignored, especially if the engine mounts provide good vibration isolation of the engine from the passenger compartment. Above that displacement, objectionable noise, vibration, and harshness (NVH) may be heard and felt by the passengers at certain engine speeds where the frequency of the engine's second harmonic coincides with one of the natural frequencies of the body structure. Then some balancing is needed in the engine to avoid customer dissatisfaction.

The shaking force for an inline engine is given by equation 14.2d (p. 647). Taking only the second harmonic term and applying the relevant factors from Table 14-4 (p. 664) for this engine gives

$$\mathbf{F}_{s_2} \cong m_B r \omega^2 \frac{4r}{l} \cos 2\omega t \hat{\mathbf{i}} \quad (14.14)$$

The shaking torque for an inline engine is given by equation 14.4c (p. 649) in combination with equation 13.15f (p. 621). Taking only the second harmonic term and applying the relevant factors from Table 14-4 for this engine gives

$$\mathbf{T}_{s_2} \cong 2m_B r^2 \omega^2 \sin 2\omega t \hat{\mathbf{k}} \quad (14.15)$$

The principle of the Lanchester balancer, shown in Figure 14-26a, can be used to counteract the secondary forces by driving the two counterrotating balance shafts at twice crankshaft speed with chains and/or gears. Figure 14-29 shows such an arrangement as applied to a Mitsubishi 2.6L, four-cylinder engine.*

H. Nakamura^[1] improved on Lanchester's 1913 design by arranging the balance shafts within the engine so as to cancel the second harmonic of the inertia torque as well as the secondary inertia force. But, his arrangement does not affect the unbalanced secondary shaking moment. In fact, it is designed to impart zero net moment about a transverse axis to either balance shaft in order to minimize bending moments on the shafts, and so reduce bearing loads and friction losses. This feature is the subject and principal claim of Nakamura's patent on this design.^[2]

Figure 14-30a shows a schematic of a conventional Lanchester balancer arranged with the two counterrotating balance shafts with their centers in a single horizontal plane transverse to the vertical plane of piston motion.

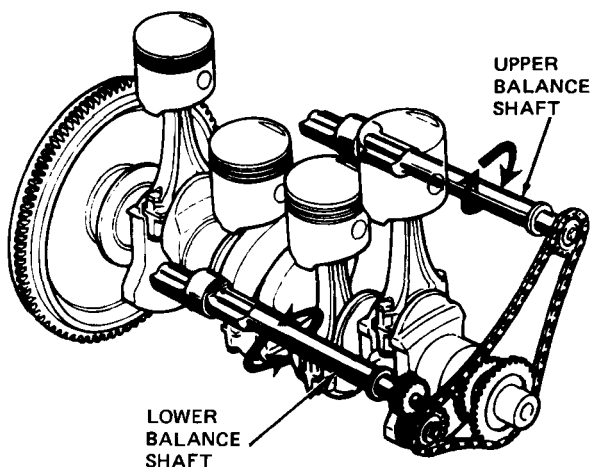
The balance force from the two balance shafts combined is

$$\mathbf{F}_{bal} \cong -8m_{bal}r_{bal}\omega^2 \cos 2\omega t \hat{\mathbf{i}} \quad (14.16)$$

where m_{bal} and r_{bal} are the mass and radius of one balance weight.

Figure 14-30b shows Nakamura's arrangement of the balance shafts with one situated above the other in separate horizontal planes. The vertical offset $x_1 - x_2$ between the shafts, in combination with the oppositely directed but equal magnitude horizontal

* Also Chrysler and Porsche under license from Mitsubishi.

**FIGURE 14-29**

Balance shafts used to eliminate the secondary unbalance in the four-cylinder inline engine
Courtesy of the Chrysler Corporation

components of the counterweights' centrifugal forces, creates a time-varying couple about the crankshaft axis defined as:

$$\mathbf{T}_{bal} \approx -4m_{bal}r_{bal}\omega^2(x_1 - x_2)\sin 2\omega t - (y_1 + y_2)\cos 2\omega t \hat{\mathbf{k}} \quad (14.17)$$

where x and y refer to the coordinates of the shaft centers referenced to the crankshaft center, and the subscripts 1 and 2 refer, respectively, to the balance shaft turning in the same and opposite directions as the crankshaft.

The vertical components of the balance weights' centrifugal forces still add to provide force balance as in equation 14.16. The torque in equation 14.17 will have opposite sense to the shaking torque if the upper shaft turns in the same direction, and the lower shaft in the opposite direction, of the crankshaft.

For force balance, equations 14.14 and 14.16 must sum to zero,

$$m_B r \omega^2 \frac{4r}{l} \cos 2\omega t - 8m_{bal}r_{bal}\omega^2 \cos 2\omega t = 0$$

or

$$m_{bal}r_{bal} = \frac{r}{2l}m_B r \quad (14.18)$$

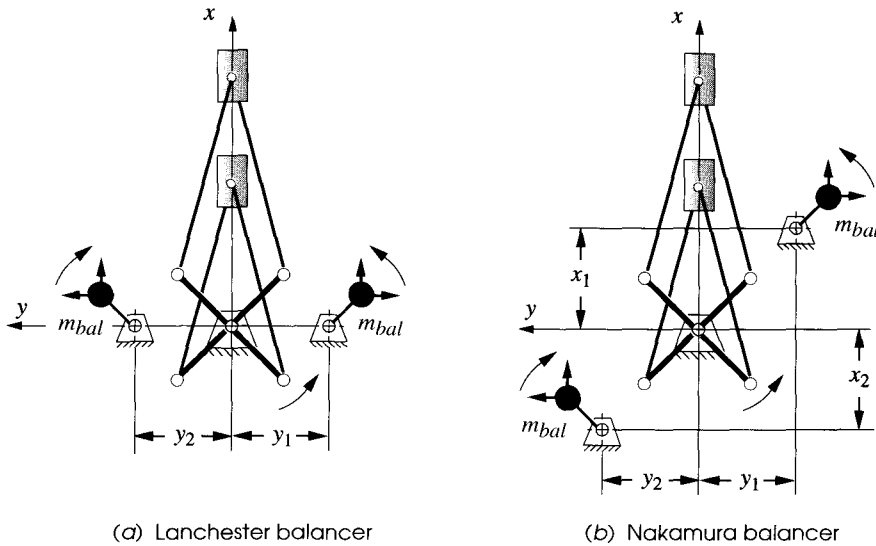
which defines the mass radius product needed for the balance mechanism.

For torque balance, equations 14.15 and 14.17 must sum to zero.

$$2m_B r^2 \omega^2 \sin 2\omega t - 4m_{bal}r_{bal}\omega^2(x_1 - x_2)\sin 2\omega t - (y_1 + y_2)\cos 2\omega t = 0 \quad (14.19a)$$

Substitute equation 14.18 in 14.19a.

$$2m_B r^2 \sin 2\omega t - 4 \frac{r}{2l} m_B r (x_1 - x_2) \sin 2\omega t - (y_1 + y_2) \cos 2\omega t = 0 \quad (14.19b)$$



(a) Lanchester balancer

(b) Nakamura balancer

FIGURE 14-30

Two types of secondary balancer mechanisms for the four-cylinder inline engine

For this equation to be zero for all ωt ,

$$\begin{aligned} y_2 &= -y_1 \\ x_1 - x_2 &= l \end{aligned} \quad (14.19c)$$

So, if the balance shafts are arranged symmetrically with respect to the piston plane at any convenient locations y_1 and $-y_1$, and the distance $x_1 - x_2$ is made equal to the length of the connecting rod l , then the second harmonic of the inertia torque will be completely cancelled. Since the second harmonic is the only nonzero component of inertia torque in this engine as can be seen in Figure 14-18b (p. 662), it will now be completely balanced for shaking force and shaking torque (but not shaking moment).

There is also significant oscillation of the gas torque in a four cylinder engine as shown in Figure 14-18a. The gas torque is 180° out of phase with the inertia torque as can be seen in Figure 14-18b and so provides some natural cancellation as shown in the total torque curve of Figure 14-18c. The magnitude of the gas torque varies with engine load and so cannot itself be cancelled with any particular balance shaft geometry for all conditions. However, one engine speed and load condition can be selected as representative of the majority of typical driving conditions, and the balance system geometry altered to give an optimum reduction of total engine torque under those conditions. Nakamura estimates that gas torque magnitude is about 30% of the inertia torque under typical driving conditions and so suggests a value of $x_1 - x_2 = 0.7l$ for the best overall reduction of total torque oscillation in this engine. Note that the average value of the driving torque is not affected by balancing because the average torque of any rotating balance system is always zero.

14.10 REFERENCES

- 1 **Nakamura, H.** (1976). "A Low Vibration Engine with Unique Counter-Balance Shafts." SAE Paper: 760111.
- 2 **Nakamura, H., et al.** (1977). "Engine Balancer." Assignee: Mitsubishi Corp.: U.S. Patent 4,028,963.

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14.12 PROBLEMS

- 14-1 Draw a crank phase diagram for a three-cylinder inline engine with a 0, 120, 240° crankshaft and determine all possible firing orders for:
 - a. Four-stroke cycle
 - b. Two-stroke cycle
 Select the best arrangement to give even firing for each stroke cycle.
- 14-2 Repeat Problem 14-1 for an inline four-cylinder engine with a 0, 90, 270, 180° crankshaft.
- 14-3 Repeat Problem 14-1 for a 45° vee, four-cylinder engine with a 0, 90, 270, 180° crankshaft.
- 14-4 Repeat Problem 14-1 for a 45° vee, two-cylinder engine with a 0, 90° crankshaft.
- 14-5 Repeat Problem 14-1 for a 90° vee, two-cylinder engine with a 0, 180° crankshaft.
- 14-6 Repeat Problem 14-1 for a 180° opposed, two-cylinder engine with a 0, 180° crankshaft.
- 14-7 Repeat Problem 14-1 for a 180° opposed, four-cylinder engine with a 0, 180, 180, 0° crankshaft.
- †14-8 Calculate the shaking force, torque, and moment balance conditions through the second harmonic for the engine design in Problem 14-1.
- †14-9 Calculate the shaking force, torque, and moment balance conditions through the second harmonic for the engine design in Problem 14-2.
- †14-10 Calculate the shaking force, torque, and moment balance conditions through the second harmonic for the engine design in Problem 14-3.
- †14-11 Calculate the shaking force, torque, and moment balance conditions through the second harmonic for the engine design in Problem 14-4.
- †14-12 Calculate the shaking force, torque, and moment balance conditions through the second harmonic for the engine design in Problem 14-5.

† These problems are suited to solution using *Mathcad*, *Matlab*, or *TKSolver* equation solver programs.

- †14-13 Calculate the shaking force, torque, and moment balance conditions through the second harmonic for the engine design in Problem 14-6.
- †14-14 Calculate the shaking force, torque, and moment balance conditions through the second harmonic for the engine design in Problem 14-7.
- 14-15 Derive expressions, in general terms, for the magnitude and angular location with respect to the first crank throw, of the mass-radius products needed on the crankshaft to balance the shaking moment in a 90° vee-eight engine with a 0, 90, 270, 180° crankshaft.
- 14-16 Repeat Problem 14-15 for a 90° vee six with a 0, 240, 120° crankshaft.
- 14-17 Repeat Problem 14-15 for a 90° vee four with a 0, 180° crankshaft.
- †14-18 Design a pair of Nakamura balance shafts to cancel the shaking force and reduce torque oscillations in the engine shown in Figure 14-18 (p. 662).*

14.13 PROJECTS

These are loosely structured design problems intended for solution using program ENGINE. All involve the design of one or more multicylinder engines and differ mainly in the specific data for the engine. The general problem statement is:

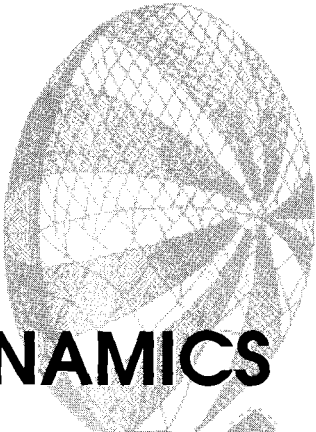
Design a multicylinder engine for a specified displacement and stroke cycle. Optimize the conrod/crank ratio and bore/stroke ratio to minimize shaking forces, shaking torque, and pin forces, also considering package size. Design your link shapes and calculate realistic dynamic parameters (mass, CG location, moment of inertia) for those links using the methods shown in Chapters 10 through 13. Dynamically model the links as described in those chapters. Balance or overbalance the linkage as needed to achieve the desired results. Choose crankshaft phase angles to optimize the inertial balance of the engine. Choose a firing order and determine the power stroke angles to optimize even firing. Trade off inertia balance if necessary to achieve even firing. Choose an optimum vee angle if appropriate to the particular problem. Overall smoothness of total torque is desired. Design and size a minimum weight flywheel by the method of Chapter 11 to smooth the total torque. Write an engineering report on your design and analysis.

- PI4-I Two-stroke cycle inline twin with a displacement of 1 liter.
- P14-2 Four-stroke cycle inline twin with a displacement of 1 liter.
- P14-3 Two-stroke cycle vee twin with a displacement of 1 liter.
- P14-4 Four-stroke cycle vee twin with a displacement of 1 liter.
- P14-5 Two-stroke cycle opposed twin with a displacement of 1 liter.
- P14-6 Four-stroke cycle opposed twin with a displacement of 1 liter.
- P14-7 Two-stroke cycle vee four with a displacement of 2 liters.
- P14-8 Four-stroke cycle vee four with a displacement of 2 liters.
- P14-9 Two-stroke cycle opposed four with a displacement of 2 liters.
- P14-10 Four-stroke cycle opposed four with a displacement of 2 liters.
- PI4-II Two-stroke inline five cylinder with a displacement of 2.5 liters.
- P14-12 Four-stroke inline five cylinder with a displacement of 2.5 liters.

† These problems are suited to solution using *Mathcad*, *Matlab*, or *TKSolver* equation solver programs.

* More information on this engine design can be found in program ENGINE where it is one of the examples.

- P14-13 Two-stroke cycle vee six with a displacement of 3 liters.
- P14-14 Four-stroke cycle vee six with a displacement of 3 liters.
- P14-15 Two-stroke cycle opposed six with a displacement of 3 liters.
- P14-16 Four-stroke cycle opposed six with a displacement of 3 liters.
- P14-17 Two-stroke inline seven cylinder with a displacement of 3.5 liters.
- P14-18 Four-stroke inline seven cylinder with a displacement of 3.5 liters.
- P14-19 Two-stroke inline eight cylinder with a displacement of 4 liters.
- P14-20 Four-stroke inline eight cylinder with a displacement of 4 liters.
- P14-21 Two-stroke vee ten cylinder with a displacement of 5 liters.
- P14-22 Four-stroke vee ten cylinder with a displacement of 5 liters.
- P14-23 Four-stroke W-6 with a displacement of 5 liters. (A W engine has three banks of cylinders on a common crank.)
- P14-24 Four-stroke W-9 with a displacement of 5 liters. (A W engine has three banks of cylinders on a common crank.)
- P14-25 Four-stroke W-12 with a displacement of 5 liters. (A W engine has three banks of cylinders on a common crank.)
- P14-26 Design a family of vee engines to be built on the same assembly line. They must all have the same vee angle and use the same pistons, connecting rods, and strokes. Their crankshafts can each be different. Four configurations are needed: vee four, vee six, vee eight, and vee ten. All will have the same single-cylinder displacement of 0.5 liters. Optimize the single-cylinder configuration from which the multicylinder engines will be constructed for bore/stroke ratio and conrod crank ratio. Then assemble this cylinder design into the above configurations. Find the best compromise of vee angle to provide a good mix of balance and even firing in all engines.
- P14-27 Repeat Project P14-26 for a family of W engines: W-3, W-6, W-9, and W-12. The inter-bank angles must be the same for all models. See the built-in example W-12 engine in program ENGINE for more information on this unusual W configuration.
- P14-28 In recent years some automobile manufacturers have made unusual vee configurations such as the VW-Audi VR15 which is a 15° vee six. Obtain detailed information on this engine design and then analyze it with program ENGINE. Write a report that explains why the manufacturer chose this unusual arrangement and justify your conclusions with sound engineering analysis.
- P14-29 Design an inline six- and an inline five-cylinder engine of the same displacement, say 2.5 liters. Analyze their dynamics with program ENGINE. Write an engineering report to explain why such manufacturers as Audi, Volvo, and Acura have chosen a five-cylinder inline over a six cylinder of comparable torque and power output.
- P14-30 Ferrari has produced vee-twelve engines in both 60° vee and horizontally opposed configurations. Design 3-liter versions of each and compare their dynamics. Write a report that explains why the manufacturer chose these arrangements and justify your conclusions with sound engineering analysis.
- P14-31 Design and compare a 3-liter 90° vee six, 60° vee six, inline six, and 180° opposed six, all of which designs are in volume production. Explain their advantages and disadvantages and justify your conclusions with sound engineering analysis.



Chapter 15

CAM DYNAMICS

*The universe is full of magical things
patiently waiting for our wits to grow sharper*
EDEN PHILLPOTS

15.0 INTRODUCTION.

Chapter 8 presented the kinematics of cams and followers and methods for their design. We will now extend the study of cam-follower systems to include considerations of the dynamic forces and torques developed. While the discussion in this chapter is limited to examples of cams and followers, the principles and approaches presented are applicable to most dynamic systems. The cam-follower system can be considered a useful and convenient example for the presentation of topics such as creating lumped parameter dynamic models and defining equivalent systems as described in Chapter 10. These techniques as well as the discussion of natural frequencies, effects of damping, and analogies between physical systems will be found useful in the analysis of all dynamic systems regardless of type.

In Chapter 10 we discussed the two approaches to dynamic analysis, commonly called the forward and the inverse dynamics problems. The forward problem assumes that all the forces acting on the system are known and seeks to solve for the resulting displacements, velocities, and accelerations. The inverse problem is, as its name says, the inverse of the other. The displacements, velocities, and accelerations are known, and we solve for the dynamic forces that result. In this chapter we will explore the application of both methods to cam-follower dynamics. Section 15.1 explores the forward solution. Section 15.3 will present the inverse solution. Both are instructive in this application of a force-closed (spring-loaded) cam-follower system and will each be discussed in this chapter.

TABLE 15-1 Notation Used in This Chapter

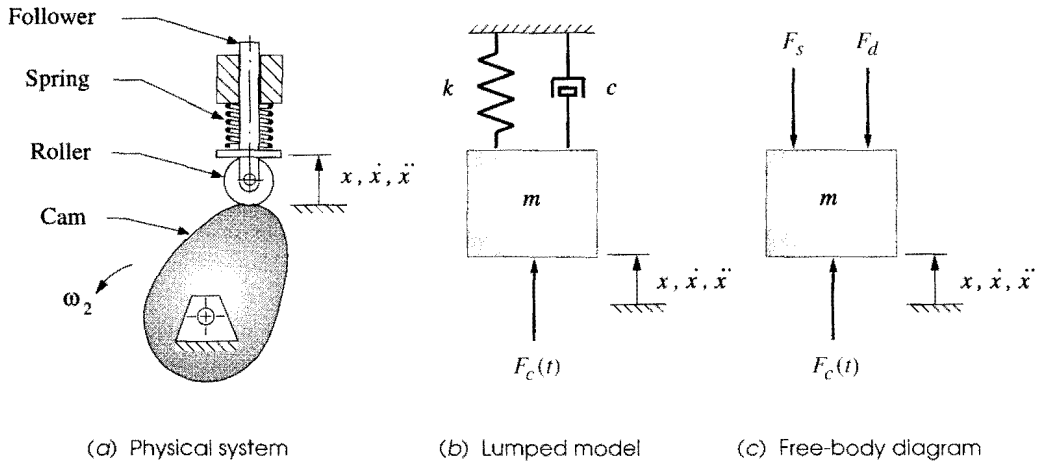
c	= damping coefficient
c_c	= critical damping coefficient
k	= spring constant
F_c	= force of cam on follower
F_s	= force of spring on follower
F_d	= force of damper on follower
m	= mass of moving elements
t	= time in seconds
T_c	= torque on camshaft
θ	= camshaft angle, in degrees or radians
ω	= camshaft angular velocity, rad/sec
ω_d	= damped circular natural frequency, rad/sec
ω_f	= forcing frequency, rad/sec
ω_n	= undamped circular natural frequency, rad/sec
x	= follower displacement, length units
$\dot{x} = v$	= follower velocity, length/sec
$\ddot{x} = a$	= follower acceleration, length/sec ²
ζ	= damping ratio

15.1 DYNAMIC FORCE ANALYSIS OF THE FORCE-CLOSED CAM-FOLLOWER

Figure 15-1a shows a simple plate or disk cam driving a spring-loaded, roller follower. This is a force-closed system which depends on the spring force to keep the cam and follower in contact at all times. Figure 15-1b shows a lumped parameter model of this system in which all the mass which moves with the follower train is lumped together as m , all the springiness in the system is lumped within the **spring constant** k , and all the **damping** or resistance to movement is lumped together as a damper with coefficient c . The sources of mass which contribute to m are fairly obvious. The mass of the follower stem, the roller, its pivot pin, and any other hardware attached to the moving assembly all add together to create m . Figure 15-1c shows the free-body diagram of the system acted upon by the cam force F_c , the spring force F_s , and the damping force F_d . There will of course also be the effects of mass times acceleration on the system.

Undamped Response

Figure 15-2 shows an even simpler lumped parameter model of the same system as in Figure 15-1 but which omits the damping altogether. This is referred to as a *conservative model* since it conserves energy with no losses. This is not a realistic or safe assumption in this case but will serve a purpose in the path to a better model which will include the damping. The free-body diagram for this mass-spring model is shown in Figure 15-2c. We can write Newton's equation for this *one-DOF* system:

**FIGURE 15-1**

One-DOF lumped parameter model of a cam-follower system including damping

$$\sum F = ma = m\ddot{x}$$

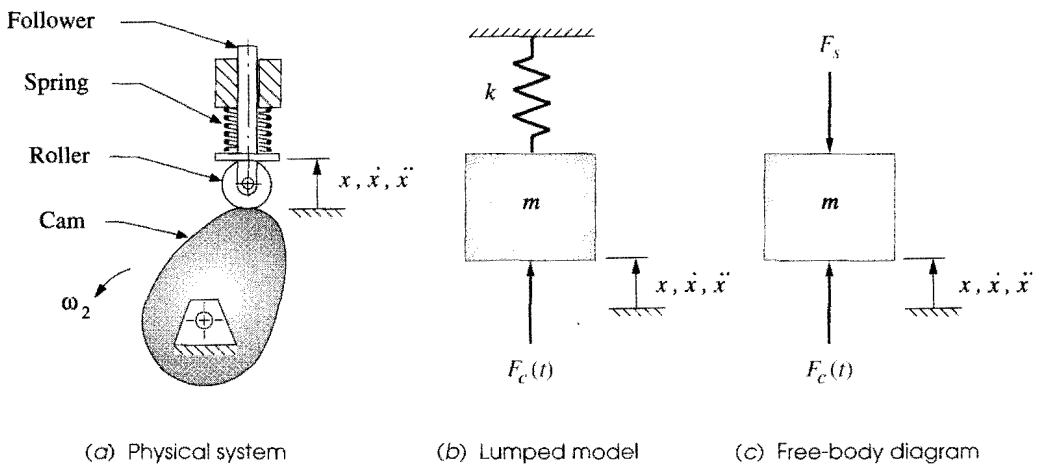
$$F_c(t) - F_s = m\ddot{x}$$

From equation 10.14:

$$F_s = kx$$

then:

$$m\ddot{x} + kx = F_c(t) \quad (15.1a)$$

**FIGURE 15-2**

One-DOF lumped parameter model of a cam-follower system without damping

This is a second-order ordinary differential equation (ODE) with constant coefficients. The complete solution will consist of the sum of two parts, the transient (homogeneous) and the steady state (particular). The homogeneous ODE is,

$$m\ddot{x} + kx = 0 \quad (15.1b)$$

$$\ddot{x} = -\frac{k}{m}x$$

which has the well-known solution,

$$x = A \cos \omega t + B \sin \omega t \quad (15.1c)$$

where A and B are the constants of integration to be determined by the initial conditions. To check the solution, differentiate it twice, assuming constant ω , and substitute in the homogeneous ODE.

$$-\omega^2 (A \cos \omega t + B \sin \omega t) = -\frac{k}{m} (A \cos \omega t + B \sin \omega t)$$

This is a solution provided that :

$$\omega^2 = \frac{k}{m} \quad \omega_n = \sqrt{\frac{k}{m}} \quad (15.1d)$$

The quantity ω_n (rad/sec) is called the *circular natural frequency* of the system and is the frequency at which the system wants to vibrate if left to its own devices. This represents the *undamped natural frequency* since we ignored damping. The *damped natural frequency* will be slightly lower than this value. Note that ω_n is a function only of the physical parameters of the system m and k ; thus it is completely determined and unchanging with time once the system is built. By creating a one-*DOF* model of the system, we have limited ourselves to one natural frequency which is an “average” natural frequency usually close to the lowest, or fundamental, frequency.

Any real physical system will also have higher natural frequencies which in general will not be integer multiples of the fundamental. In order to find them we need to create a multi-degree-of-freedom model of the system. The fundamental tone at which a bell rings when struck is its natural frequency defined by this expression. The bell also has overtones which are its other, higher, natural frequencies. The fundamental frequency tends to dominate the transient response of the system.^[1]

The circular natural frequency ω_n (rad/sec) can be converted to cycles per second (hertz) by noting that there are 2π radians per revolution and one revolution per cycle:

$$f_n = \frac{1}{2\pi} \omega_n \quad \text{hertz} \quad (15.1e)$$

The constants of integration, A and B in equation 15.1c, depend on the initial conditions. A general case can be stated as,

When $t = 0$, let $x = x_0$ and $v = v_0$, where x_0 and v_0 are constants

which gives a general solution to the homogeneous ODE 15.1b of:

$$x = x_0 \cos \omega_n t + \frac{v_0}{\omega_n} \sin \omega_n t \quad (15.1f)$$

Equation 15.1f can be put into polar form by computing the magnitude and phase angle:

$$X_0 = \sqrt{x_0^2 + \left(\frac{v_0}{\omega_n}\right)^2} \quad \phi = \arctan\left(\frac{v_0}{x_0 \omega_n}\right)$$

then :

$$x = X_0 \cos(\omega_n t - \phi) \quad (15.1g)$$

Note that this is a pure harmonic function whose amplitude X_0 and phase angle ϕ are a function of the initial conditions and the natural frequency of the system. It will oscillate forever in response to a single, transitory input if there is truly no damping present.

Damped Response

If we now reintroduce the damping of the model in Figure 15-1b and draw the free-body diagram as shown in Figure 15-1c, the summation of forces becomes:

$$F_c(t) - F_d - F_s = m\ddot{x} \quad (15.2a)$$

Substituting equations 10.14 (p. 501) and 10.15c (p. 503):

$$m\ddot{x} + c\dot{x} + kx = F_c(t) \quad (15.2b)$$

HOMOGENEOUS SOLUTION We again separate this differential equation into its homogeneous and particular components. The homogeneous part is:

$$\ddot{x} + \frac{c}{m}\dot{x} + \frac{k}{m}x = 0 \quad (15.2c)$$

The solution to this ODE is of the form:

$$x = Re^{st} \quad (15.2d)$$

where R and s are constants. Differentiating versus time:

$$\dot{x} = Rse^{st}$$

$$\ddot{x} = Rs^2e^{st}$$

and substituting in equation 15.2c:

$$Rs^2e^{st} + \frac{c}{m}Rse^{st} + \frac{k}{m}Re^{st} = 0$$

$$\left(s^2 + \frac{c}{m}s + \frac{k}{m}\right)Re^{st} = 0 \quad (15.2e)$$

For this solution to be valid either R or the expression in parentheses must be zero as e^{st} is never zero. If R were zero, then the assumed solution, in equation 15.2d, would also be zero and thus not be a solution. Therefore, the quadratic equation in parentheses must be zero.

$$\left(s^2 + \frac{c}{m}s + \frac{k}{m}\right) = 0 \quad (15.2f)$$

This is called the characteristic equation of the ODE and its solution is:

$$s = \frac{-\frac{c}{m} \pm \sqrt{\left(\frac{c}{m}\right)^2 - 4\frac{k}{m}}}{2}$$

which has the two roots:

$$s_1 = -\frac{c}{2m} + \sqrt{\left(\frac{c}{2m}\right)^2 - \frac{k}{m}} \quad (15.2g)$$

$$s_2 = -\frac{c}{2m} - \sqrt{\left(\frac{c}{2m}\right)^2 - \frac{k}{m}}$$

These two roots of the characteristic equation provide two independent terms of the homogeneous solution:

$$x = R_1 e^{s_1 t} + R_2 e^{s_2 t} \quad \text{for } s_1 \neq s_2 \quad (15.2h)$$

If $s_1 = s_2$, then another form of solution is needed. The quantity s_1 will equal s_2 when:

$$\sqrt{\left(\frac{c}{2m}\right)^2 - \frac{k}{m}} = 0 \quad \text{or:} \quad \frac{c}{2m} = \sqrt{\frac{k}{m}}$$

and:

$$c = 2m\sqrt{\frac{k}{m}} = 2m\omega_n = c_c \quad (15.2i)$$

This particular value of c is called the **critical damping** and is labeled c_c . The system will behave in a unique way when critically damped, and the solution must be of the form:

$$x = R_1 e^{s_1 t} + R_2 t e^{s_2 t} \quad \text{for } s_1 = s_2 = -\frac{c}{2m} \quad (15.2j)$$

It will be useful to define a dimensionless ratio called the **damping ratio** ζ which is the actual damping divided by the critical damping.

$$\zeta = \frac{c}{c_c} \quad (15.3a)$$

$$\zeta = \frac{c}{2m\omega_n}$$

and then:

$$\zeta \omega_n = \frac{c}{2m} \quad (15.3b)$$

The damped natural frequency ω_d is slightly less than the undamped natural frequency ω_n and is:

$$\omega_d = \sqrt{\frac{k}{m} - \left(\frac{c}{2m}\right)^2} \quad (15.3c)$$

We can substitute equations 15.1d and 15.3b into equations 15.2g to get an expression for the characteristic equation in terms of dimensionless ratios:

$$s_{1,2} = -\omega_n \zeta \pm \sqrt{(\omega_n \zeta)^2 - \omega_n^2} \quad (15.4a)$$

$$s_{1,2} = \omega_n \left(-\zeta \pm \sqrt{\zeta^2 - 1} \right)$$

This shows that the system response is determined by the damping ratio ζ which dictates the value of the discriminant. There are three possible cases:

CASE 1:	$\zeta > 1$	Roots real and unequal	(15.4b)
CASE 2:	$\zeta = 1$	Roots real and equal	
CASE 3:	$\zeta < 1$	Roots complex conjugate	

Let's consider the response of each of these cases separately.

CASE 1: $\zeta > 1$ *overdamped*

The solution is of the form in equation 15.2h and is:

$$x = R_1 e^{(-\zeta + \sqrt{\zeta^2 - 1}) \omega_n t} + R_2 e^{(-\zeta - \sqrt{\zeta^2 - 1}) \omega_n t} \quad (15.5a)$$

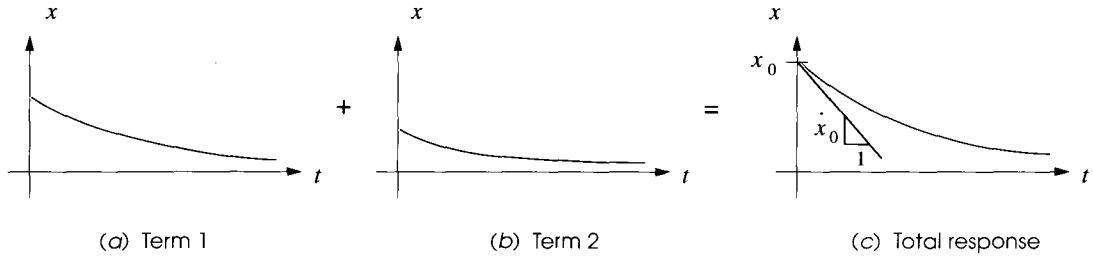
Note that since $\zeta > 1$, both exponents will be negative making x the sum of two decaying exponentials as shown in Figure 15-3. This is the transient response of the system to a disturbance and dies out after a time. There is no oscillation in the output motion. An example of an overdamped system which you have probably encountered is the tone arm on a good-quality record turntable with a "cueing" feature. The tone arm can be lifted up, then released, and it will slowly "float" down to the record. This is achieved by putting a large amount of damping in the system, at the arm pivot. The arm's motion follows an exponential decay curve such as in Figure 15-3.

CASE 2: $\zeta = 1$ *critically damped*

The solution is of the form in equation 15.2j and is:

$$x = R_1 e^{-\omega_n t} + R_2 t e^{-\omega_n t} = (R_1 + R_2 t) e^{-\omega_n t} \quad (15.5b)$$

This is the product of a linear function of time and a decaying exponential function and can take several forms depending on the values of the constants of integration, R_1

**FIGURE 15-3**

Transient response of an overdamped system

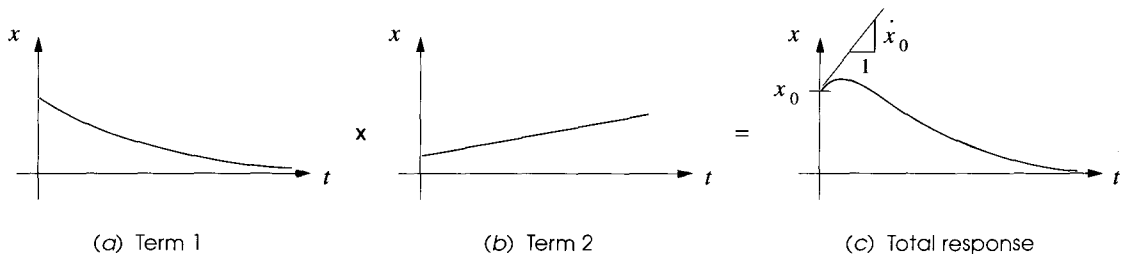
and R_2 , which in turn depend on initial conditions. A typical transient response might look like Figure 15-4. This is the transient response of the system to a disturbance, which response dies out after a time. There is fast response but no oscillation in the output motion. An example of a critically damped system is the suspension system of a new sports car in which the damping is usually made close to critical in order to provide crisp handling response without either oscillating or being slow to respond. A critically damped system will, when disturbed, return to its original position within one bounce. It may overshoot but will not oscillate and will not be sluggish.

CASE 3: $\zeta < 1$ underdamped

The solution is of the form in equation 15.2h and s_1, s_2 are complex conjugate. Equation 15.4a can be rewritten in a more convenient form as:

$$s_{1,2} = \omega_n \left(-\zeta \pm j\sqrt{1-\zeta^2} \right); \quad j = \sqrt{-1} \quad (15.5c)$$

Substituting in equation 15.2h:

**FIGURE 15-4**

Transient response of a critically damped system

$$x = R_1 e^{(-\zeta + j\sqrt{1-\zeta^2})\omega_n t} + R_2 e^{(-\zeta - j\sqrt{1-\zeta^2})\omega_n t}$$

and noting that :

$$y^{a+b} = y^a y^b$$

$$x = R_1 \left[e^{-\zeta\omega_n t} e^{(j\sqrt{1-\zeta^2})\omega_n t} \right] + R_2 \left[e^{-\zeta\omega_n t} e^{(-j\sqrt{1-\zeta^2})\omega_n t} \right]$$

factor :

$$x = e^{-\zeta\omega_n t} \left[R_1 e^{(j\sqrt{1-\zeta^2})\omega_n t} + R_2 e^{(-j\sqrt{1-\zeta^2})\omega_n t} \right] \quad (15.5d)$$

Substitute the Euler identity from equation 4.4a (p. 155):

$$x = e^{-\zeta\omega_n t} \left\{ \begin{aligned} &R_1 \left[\cos(\sqrt{1-\zeta^2}\omega_n t) + j \sin(\sqrt{1-\zeta^2}\omega_n t) \right] \\ &+ R_2 \left[\cos(\sqrt{1-\zeta^2}\omega_n t) - j \sin(\sqrt{1-\zeta^2}\omega_n t) \right] \end{aligned} \right\} \quad (15.5e)$$

and simplify :

$$x = e^{-\zeta\omega_n t} \left\{ (R_1 + R_2) \left[\cos(\sqrt{1-\zeta^2}\omega_n t) \right] + (R_1 - R_2) j \sin(\sqrt{1-\zeta^2}\omega_n t) \right\}$$

Note that R_1 and R_2 are just constants yet to be determined from the initial conditions, so their sum and difference can be denoted as some other constants:

$$x = e^{-\zeta\omega_n t} \left\{ A \left[\cos(\sqrt{1-\zeta^2}\omega_n t) \right] + B \sin(\sqrt{1-\zeta^2}\omega_n t) \right\} \quad (15.5f)$$

We can put this in polar form by defining the magnitude and phase angle as:

$$X_0 = \sqrt{A^2 + B^2} \quad \phi = \arctan \frac{B}{A} \quad (15.5g)$$

then :

$$x = X_0 e^{-\zeta\omega_n t} \cos \left[\left(\sqrt{1-\zeta^2}\omega_n t \right) - \phi \right]$$

This is the product of a harmonic function of time and a decaying exponential function where X_0 and ϕ are the constants of integration determined by the initial conditions.

Figure 15-5 shows the transient response for this **underdamped case**. The response overshoots and oscillates before finally settling down to its final position. Note that if the damping ratio ζ is zero, equation 15.5g reduces to equation 15.1g which is a pure harmonic.

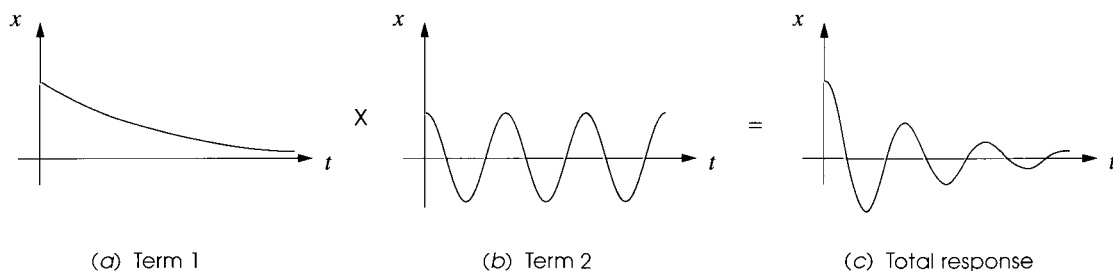


FIGURE 15-5

Transient response of an underdamped system

An example of an **underdamped system** is a diving board which continues to oscillate after the diver has jumped off, finally settling back to zero position. *Many real systems in machinery are underdamped, including the typical cam-follower system.* This often leads to **vibration problems**. It is not usually a good solution simply to add damping to the system as this causes heating and is very energy inefficient. It is better to design the system to avoid the vibration problems.

PARTICULAR SOLUTION Unlike the homogeneous solution which is always the same regardless of the input, the particular solution to equation 15.2b (p. 689) will depend on the forcing function $F_c(t)$ which is applied to the cam-follower from the cam. In general the output displacement x of the follower will be a function of similar shape to the input function but will lag the input function by some phase angle. It is quite reasonable to use a sinusoidal function as an example since any periodic function can be represented as a Fourier series of sine and cosine terms of different frequencies (see equation 13.2, p. 608, equation 13.3, p. 609, and the footnote on p. 609).

Assume the forcing function to be:

$$F_c(t) = F_0 \sin \omega_f t \quad (15.6a)$$

where F_0 is the amplitude of the force and ω_f is its circular frequency. Note that ω_f is unrelated to ω_n or ω_d and may be any value. The system equation then becomes:

$$m\ddot{x} + c\dot{x} + kx = F_0 \sin \omega_f t \quad (15.6b)$$

The solution must be of harmonic form to match this forcing function and we can try the same form of solution as used for the homogeneous solution.

$$x_f(t) = X_f \sin(\omega_f t - \psi) \quad (15.6c)$$

where:

X_f = amplitude

ψ = phase angle between applied force and displacement

ω_f = angular velocity of forcing function

The factors X_f and ψ are not constants of integration here. They are constants determined by the physical characteristics of the system and the forcing function's frequency and magnitude. They have nothing to do with the initial conditions. To find their values, differentiate the assumed solution twice, substitute in the ODE, and get:

$$X_f = \frac{F_0}{\sqrt{(k - m\omega_f^2)^2 + (c\omega_f)^2}} \quad (15.6d)$$

$$\psi = \arctan \left[\frac{c\omega_f}{(k - m\omega_f^2)^2} \right]$$

Substitute equations 15.1d, 15.2i, and 15.3a and put in dimensionless form:

$$\frac{X_f}{\left(\frac{F_0}{k}\right)} = \frac{1}{\sqrt{\left[1 - \left(\frac{\omega_f}{\omega_n}\right)^2\right]^2 + \left(2\zeta \frac{\omega_f}{\omega_n}\right)^2}} \quad (15.6e)$$

$$\psi = \arctan \left[\frac{2\zeta \frac{\omega_f}{\omega_n}}{1 - \left(\frac{\omega_f}{\omega_n}\right)^2} \right]$$

The ratio ω_f / ω_n is called the **frequency ratio**. Dividing X_f by the static deflection F_0 / k creates the **amplitude ratio** which defines the relative dynamic displacement compared to the static.

COMPLETE RESPONSE The complete solution to our system differential equation for a sinusoidal forcing function is the sum of the homogeneous and particular solutions:

$$x = X_0 e^{-\zeta \omega_n t} \cos \left[\left(\sqrt{1 - \zeta^2} \omega_n t \right) - \phi \right] + X_f \sin(\omega_f t - \psi) \quad (15.7)$$

The homogeneous term represents the **transient response** of the system which will die out in time but is reintroduced any time the system is again disturbed. The **particular term** represents the **forced response** or **steady-state response** to a sinusoidal forcing function which will continue as long as the forcing function is present.

Note that the solution to this equation, shown in equations 15.5 and 15.6, depends only on two ratios, the damping ratio ζ which relates the actual damping relative to the critical damping, and the *frequency ratio* ω_f / ω_n which relates the forcing frequency to the natural frequency of the system. Koster^[1] found that a typical value for the damping ratio in cam-follower systems is $\zeta = 0.06$, so they are underdamped and can **resonate** if operated at frequency ratios close to 1.

The initial conditions for the specific problem are applied to equation 15.7 to determine the values of X_0 and ϕ . Note that these constants of integration are contained within the homogeneous part of the solution.

15.2 RESONANCE

The natural frequency (and its overtones) are of great interest to the designer as they define the frequencies at which the system will **resonate**. The single-DOF lumped parameter systems shown in Figures 15-1 and 15-2 (p. 687) are the simplest possible to describe a dynamic system, yet they contain all the basic dynamic elements. Masses and springs are energy storage elements. A mass stores kinetic energy, and a spring stores potential energy. The damper is a dissipative element. It uses energy and converts it to heat. Thus all the losses in the model of Figure 15-1 occur through the damper.

These are "pure" idealized elements which possess only their own characteristics. That is, the spring has no damping and the damper no springiness, etc. Any system which contains more than one energy storage device such as a mass and a spring will possess at least one natural frequency. If we excite the system at its natural frequency, we will set up the condition called resonance in which the energy stored in the system's elements will oscillate from one element to the other at that frequency. The result can be violent oscillations in the displacements of the movable elements in the system as the energy moves from potential to kinetic form and vice versa.

Figure 15-6 shows a plot of the amplitude and phase angle of the displacement response X of the system to a sinusoidal input forcing function at various frequencies ω . In our case, the forcing frequency is the angular velocity at which the cam is rotating. The plot normalizes the forcing frequency as the frequency ratio ω/ω_n . The response amplitude X is normalized by dividing the dynamic deflection x by the static deflection F_0/k that the same force amplitude would create on the system. Thus at a frequency of zero, the output is one, equal to the static deflection of the spring at the amplitude of the input force. As the forcing frequency increases toward the natural frequency ω_n , the amplitude of the output motion, for zero damping, increases rapidly and becomes theoretically infinite when $\omega = \omega_n$. Beyond this point the amplitude decreases rapidly and asymptotically toward zero at high frequency ratios. It is obvious that we must avoid driving this system at or near its natural frequency. The addition of damping reduces the amplitude of vibration at the natural frequency, but very large damping ratios (> 0.6) are needed to keep the output amplitude less than or equal to the input amplitude. This is much more damping than is found in cam-follower systems and most machinery. One result of operation of an underdamped cam-follower system near ω_n can be **follower jump**. The system of follower mass and spring can oscillate violently at its natural frequency and leave contact with the cam. When it does reestablish contact, it may do so with severe impact loads that can quickly fail the materials.

The designer has a degree of control over resonance in that the system's mass m and stiffness k can be tailored to move its natural frequency away from any required operating frequencies. A common rule of thumb is to design the system to have a fundamental natural frequency ω_n at least ten times the highest forcing frequency expected in service, thus keeping all operation well below the resonance point. This is often difficult to achieve in mechanical systems. One tries to achieve the largest ratio ω_n/ω possible nevertheless, even if it is less than 10.

For cams with finite jerk having a dwell after the rise of the same order of duration as the rise, Koster[1] defines a dimensionless ratio,

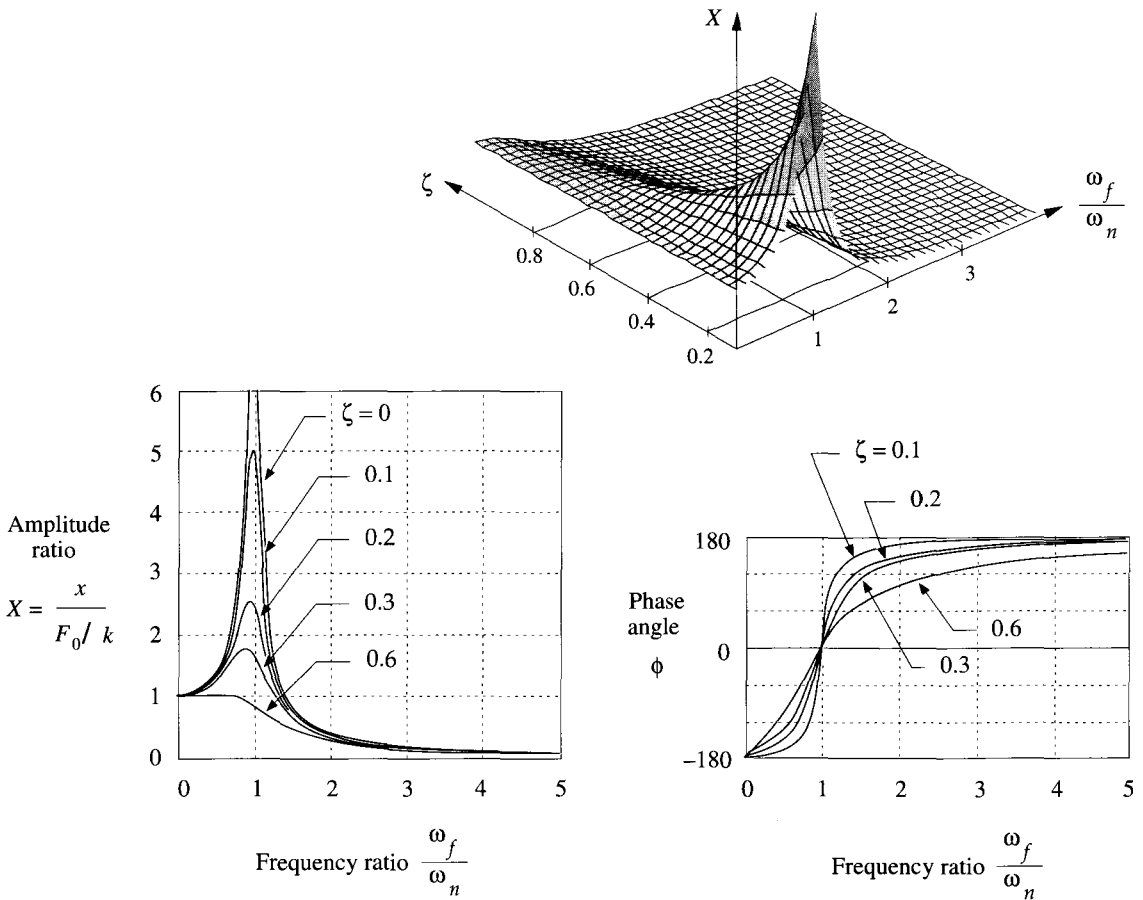


FIGURE 15-6

Amplitude ratio and phase angle of system response

$$\tau = \frac{T_n}{t_r} \qquad T_n = \frac{1}{\omega_n} \qquad (15.8)$$

where T_n is the period of the follower system's natural frequency and t_r is the time to complete the rise portion of the follower motion. To minimize the effects of the transient vibrations on the system, this ratio should be as small as possible and always less than 1. A small value of τ is equivalent to a large ratio of ω_n / ω_f . The error in acceleration of the follower due to vibrations during the dwell period will be proportional to the first power of τ , and the error in follower position will be proportional to the third power of τ , for values of $\tau < 0.5$. If the dwell's duration is about as long or longer than that of the rise, these transient vibrations will tend to die out by the end of the dwell. The next fall or rise will again provide an input to the system and cause a new transient response.

Thus, the response of a cam-follower system of this type will be dominated by the recurring transient responses rather than by the forced, or steady-state, response. It is important to adhere to the fundamental law of cam design and use cam programs with finite jerk in order to minimize these residual vibrations in the follower system. Koster[1] reports that the cycloidal and 3-4-5 polynomial programs both gave low residual vibrations in the double-dwell cam. The modified sine acceleration program will also give good results. All these have finite jerk. (See Section 8.8 and Table 8-4, p. 421.)

Some thought and observation of equation 15.1d (p. 688) will show that we would like our system members to be both light (low m) and stiff (high k) to get high values for ω_n and thus low values for ω/ω_n . Unfortunately, the lightest materials are seldom also the stiffest. Aluminum is one-third the weight of steel but is also about a third as stiff. Titanium is about half the weight of steel but also about half as stiff. Some of the exotic composite materials such as carbon fiber/epoxy offer better stiffness-to-weight ratios but their cost is high and processing is difficult. Another job for *Unobtainium* 208!

Note in Figure 15-6 (p. 697) that the amplitude of vibration at large frequency ratios approaches zero. So, if the system can be brought up to speed through the resonance point without damage and then kept operating at a large frequency ratio, the vibration will be minimal. An example of systems designed to be run this way are large devices that must run at higher speed such as electrical power generators. Their large mass creates a lower natural frequency than their required operating speeds. They are "run up" as quickly as possible through the resonance region to avoid damage from their vibrations and "rundown" quickly through resonance when stopping them. They also have the advantage of long duty cycles of constant speed operation in the safe frequency region between infrequent starts and stops.

15.3 KINETOSTATIC FORCE ANALYSIS OF THE FORCE-CLOSED CAM-FOLLOWER

The analysis in the previous sections is of the "**forward dynamic analysis**" type as defined in Section 10.1, p. 491. That is, the applied force $[F_e(t)]$ in equation 15.2b, p. 689] is presumed to be known, and the equation is solved for the resulting displacement x from which its derivatives can also be determined. Cam-follower force analysis problems, like linkage analysis problems, lend themselves well to solution by the **inverse dynamics**, or **kinetostatics**, approach. Here the displacement and its derivatives are defined from the kinematic design of the cam based on an assumed angular velocity ω of the cam, and we want to solve for the force $F_e(t)$. Equation 15.2b can be solved directly for the force in a spring-loaded cam-follower system provided that values for mass m , spring constant k , preload F_{pz} , and damping factor c are known in addition to the displacement, velocity, and acceleration functions.

The designer has a large degree of control over the system spring constant k_{eff} as it tends to be dominated by the k s of the physical return spring. The elasticities of the follower parts also contribute to the overall system k_{eff} but are usually much stiffer than the physical spring. If the follower stiffness is in series with the return spring, as it often is, equation 10.17c (p. 506) shows that the softest spring in series will dominate the effective spring constant. Thus the return spring will virtually determine the overall k unless some parts of the follower train have similarly low stiffness.

The designer will choose or design the return spring and thus can specify both its k and the amount of preload to be introduced at assembly. Preload of a spring occurs when it is compressed (or extended if an extension spring) from its *free length* to its initial assembled length. This is a necessary and desirable situation as we want some residual force on the follower even when the cam is at its lowest displacement. This will help maintain good contact between the cam and follower at all times. This spring preload F_{pl} adds a constant term to equation 15.2b which becomes:

$$F_c(t) = m\ddot{x} + c\dot{x} + kx + F_{pl} \quad (15.9)$$

The value of m is determined from the effective mass of the system as lumped in the single-DOF model of Figure 15-1 (p. 687). The value of c for most cam-follower systems can be estimated for a first approximation to be about 0.05 to 0.15 of the critical damping c_c as defined in equation 15.2i (p. 690). Koster^[1] found that a typical value for the damping ratio in cam-follower systems is $\zeta = 0.06$.

Calculating the damping c based on an assumed value of ζ requires specifying a value for the overall system k and for its effective mass. The choice of k will affect both the natural frequency of the system for a given mass and the available force to keep the joint closed. Some iteration will probably be needed to find a good compromise. A selection of data for commercially available helical coil springs is provided in Appendix D. Note in equation 15.9 that the terms involving acceleration and velocity can be either positive or negative. The terms involving the spring parameters k and F_{pl} are the only ones that are always positive. So, to keep the overall function always positive requires that the spring force terms be large enough to counteract any negative values in the other terms. Typically, the acceleration is larger numerically than the velocity, so the negative acceleration usually is the principal cause of a negative force F_c .

The principal concern in this analysis is to keep the cam force always positive in sign as its direction is defined in Figure 15-1. The cam force is shown as positive in that figure. In a force-closed system the cam can only push on the follower. It cannot pull. The follower spring is responsible for providing the force needed to keep the joint closed during the negative acceleration portions of the follower motion. The damping force also can contribute, but the spring must supply the bulk of the force to maintain contact between the cam and follower. If the force F_c goes negative at any time in the cycle, the follower and cam will part company, a condition called **follower jump**. When they meet again, it will be with large and potentially damaging impact forces. The follower jump, if any, will occur near the point of maximum negative acceleration. Thus we must select the spring constant and preload to guarantee a positive force at all points in the cycle. In automotive engine valve cam applications follower jump is also called *valve float*, because the valve (follower) “floats” above the cam, also periodically impacting the cam surface. This will occur if the cam rpm is increased to the point that the larger negative acceleration makes the follower force negative. The “redline” maximum engine rpm often indicated on its tachometer is to warn of impending valve float above that speed which will damage the cam and follower.

Program DYNACAM allows the iteration of equation 15.9 to be done quickly and painlessly for any cam whose kinematics have been defined in that program. The program's *Dynamics* button will solve equation 15.9 for all values of camshaft angle, using the displacement, velocity, and acceleration functions previously calculated for that cam design in the program. The program requires values for the effective system mass m ,

effective spring constant k , preload $\dot{F}_{pl}$, and the assumed value of the damping ratio ζ . These values need to be determined for the model by the designer using the methods described in Sections 10.9 (p. 500) and 10.10 (p. 503). The calculated force at the cam-follower interface can then be plotted or its values printed in tabular form. The system's natural frequency is also reported when the tabular force data are printed.



EXAMPLE 15-1

Kinetostatic Force Analysis of a Force-Closed (Spring-loaded) Cam-Follower System.

Given: A translating roller follower as shown in Figure 15-1 is driven by a force-closed radial plate cam which has the following program:

Segment 1: Rise 1 inch in 50° with modified sine acceleration
 Segment 2: Dwell for 40°
 Segment 3: Fall 1 inch in 50° with cycloidal displacement
 Segment 4: Dwell for 40°
 Segment 5: Rise 1 inch in 50° with 3-4-5 polynomial displacement
 Segment 6: Dwell for 40°
 Segment 7: Fall 1 inch in 50° with 4-5-6-7 polynomial displacement
 Segment 8: Dwell for 40°
 Camshaft angular velocity is 18.85 rad/sec.
 Follower effective mass is 0.0738 in-lb-sec² (blobs).
 Damping is 15% of critical ($\bar{S} = 0.15$).

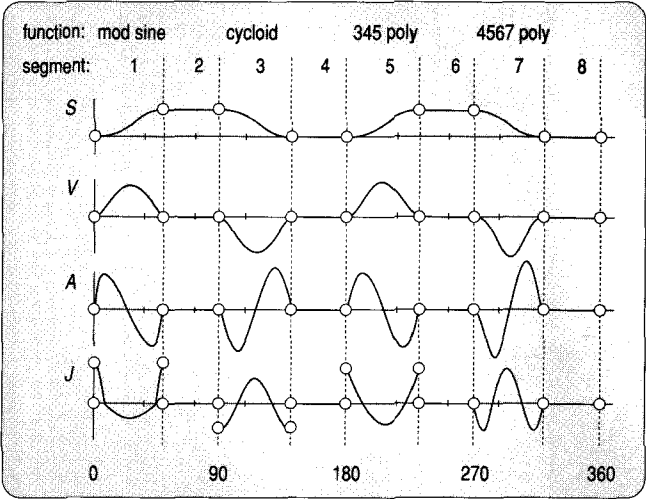
Problem: Compute the necessary spring constant and spring preload to maintain contact between cam and follower and calculate the dynamic force function for the cam. Calculate the system natural frequency with the selected spring. Keep the pressure angle under 30° .

Solution:

- 1 Calculate the kinematic data (follower displacement, velocity, acceleration, and jerk) for the specified cam functions. The acceleration for this cam is shown in Figure 15-7 and has a maximum value of 3504 in/sec². See Chapter 8 to review this procedure.
- 2 Calculate the pressure angle and radius of curvature for trial values of prime circle radius, and size the cam to control these values. Figure 15-8 shows the pressure angle function and Figure 15-9 the radii of curvature for this cam with a prime circle radius of 4 in and zero eccentricity. The maximum pressure angle is 29.2° and the minimum radius of curvature is 1.7 in. Figure 8-47 (p. 407) shows the finished cam profile. See Chapter 8 to review these calculations.
- 3 With the kinematics of the cam defined, we can address its dynamics. To solve equation 15.9 for cam force, we must assume values for the spring constant k and the preload F_{pl} . The value of c can be calculated from equation 15.3a using the given mass m , the damping factor $\bar{S}$, and assumed k . The kinematic parameters are known.
- 4 Program DYNACAM does this computation for you. The dynamic force that results from an assumed k of 150 lb/in and a preload of 75 lb is shown in Figure 15-10a. The damping coefficient $c = 0.998$. Note that the force dips below the zero axis in two places during negative acceleration. These are locations of follower jump. The follower has left the cam during the

Segment Number	Function used	Start Angle	End Angle	Delta Angle
1	ModSine rise	0	50	50
2	Dwell	50	90	40
3	Cycloid fall	90	140	50
4	Dwell	140	180	40
5	345 poly rise	180	230	50
6	Dwell	230	270	40
7	4567 poly fall	270	320	50
8	Dwell	320	360	40

(a) Cam program specifications



(b) Plots of cam-follower's S V A J diagrams

FIGURE 15-7
S V A J diagrams for Examples 15-1 and 15-2 (p. 705)

fall because the spring does not have enough available force to keep the follower in contact with the rapidly falling cam. Open the file E15-01.cam in DYNACAM and provide the specified k and F_{pl} to see this example. Another iteration is needed to improve the design.

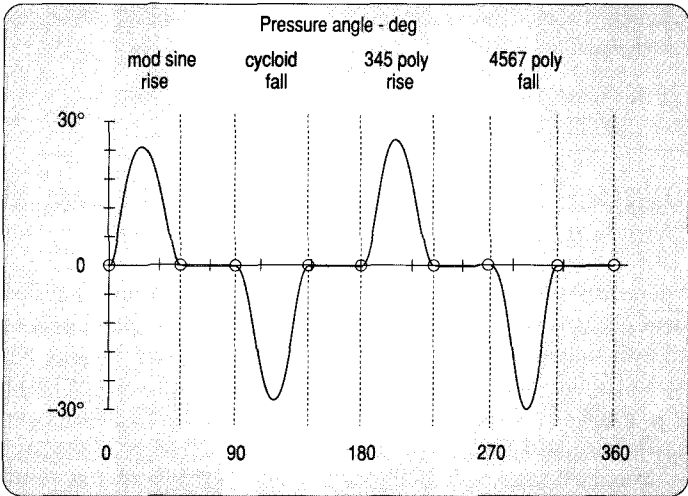
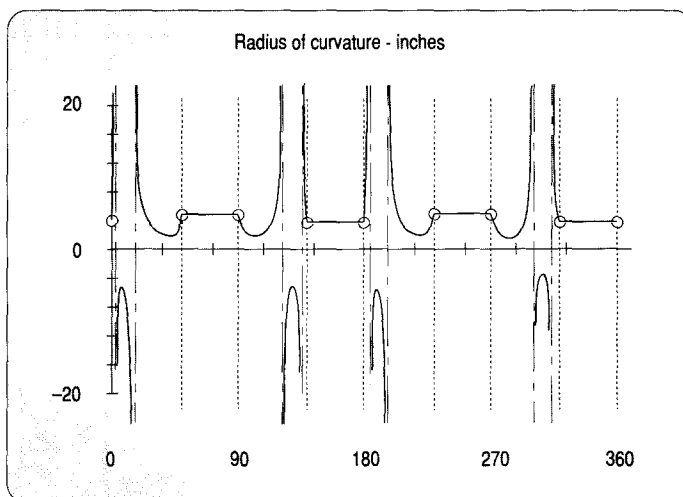


FIGURE 15-8
Pressure angle plot for Examples 15-1 and 15-2 (p. 705)

**FIGURE 15-9**

Radius of curvature of a four-dwell cam for Examples 15-1 and 15-2 (p. 705)

- 5 Figure 15-10b shows the dynamic force for the same cam with a spring constant of $k = 200$ lb/in and a preload of 150 lb. The damping coefficient $c = 1.153$. This additional force has lifted the function up sufficiently to keep it positive everywhere. There is no follower jump in this case. The maximum force during the cycle is 400.4 lb. A margin of safety has been provided by keeping the minimum force comfortably above the zero line at 36.9 lb. Run example #5 in the program and provide the specified spring constant and preload values to see this example.
- 6 The fundamental natural frequency, both undamped and damped, can be calculated for the system from equations 15.1d (p. 688) and 15.3c (p. 691) and are:

$$\omega_n = 52.06 \text{ rad/sec};$$

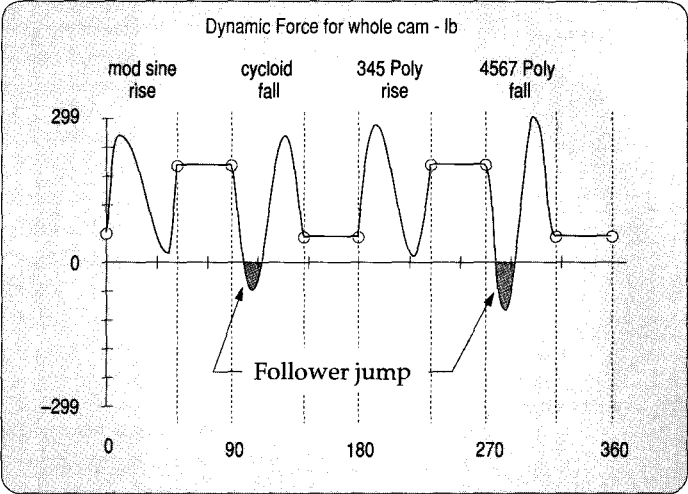
$$\omega_d = 51.98 \text{ rad/sec}$$

15.4 KINETOSTATIC FORCE ANALYSIS OF THE FORM-CLOSED CAM-FOLLOWER

Section 8.1 described two types of joint closure used in cam-follower systems, **force closure** and **form closure**. Force closure uses an open joint and requires a spring or other force source to maintain contact between the elements. Form closure provides a geometric constraint at the joint such as the cam groove shown in Figure 15-11a or the conjugate cams of Figure 15-11b. No spring is needed to keep the follower in contact with these cams. The follower will run against one side or the other of the groove or conjugate pair as necessary to provide both positive and negative forces. Since there is no spring in this system, its dynamic force equation 15.9 (p. 699) simplifies to:

$$F_c(t) = m\ddot{x} + c\dot{x} \quad (15.10)$$

(a) Insufficient spring force allows follower jump



(b) Sufficient spring force keeps the dynamic force positive

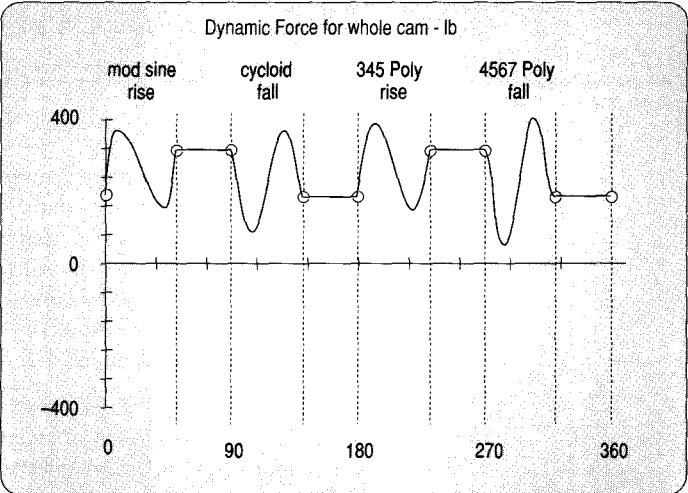
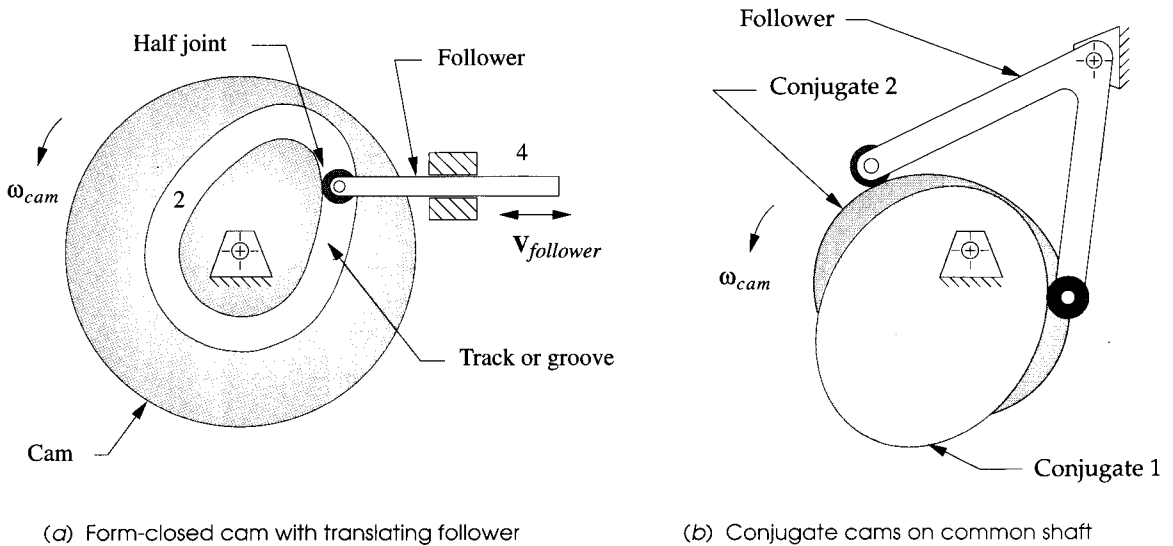


FIGURE 15-10

Dynamic forces in a force-closed cam-follower system

Note that there is now only one energy storage element in the system (the mass), so, theoretically, resonance is not possible. There is no natural frequency for it to resonate at. This is the chief advantage of a form-closed system over a force-closed one. Follower jump will not occur, short of complete failure of the parts, no matter how fast the system is run. This arrangement is sometimes used in high-performance or racing engine valve trains to allow higher redline engine speeds without valve float. In engine valve trains, a form-closed cam-follower valve train is called a *desmodromic* system.

As with any design, there are trade-offs. While the form-closed system typically allows higher operating speeds than a comparable force-closed system, it is not free of all

**FIGURE 15-11**

Form-closed cam-follower systems

vibration problems. Even though there is no physical return spring in the system, the follower train, the camshaft, and all other parts still have their own spring constants which combine to provide an overall effective spring constant for the system as shown in Section 15.2 (p. 696). The positive side is that this spring constant will typically be quite large (stiff) since properly designed follower parts are designed to be stiff. The effective natural frequency will then be high (see equation 15.1d, p. 688) and possibly well above the forcing frequency as desired.

Another problem with form-closed cams, especially the grooved or track type shown in Figure 15-11a, is the phenomenon of **crossover shock**. Every time the acceleration of the follower changes sign, the inertial force also does so. This causes the follower to abruptly shift from one side of the cam groove to the other. There cannot be zero clearance between the roller follower and the groove and still have it operate. Even if the clearance is very small, there will still be an opportunity for the follower to develop some velocity in its short trip across the groove, and it will impact the other side. Track cams of the type shown in Figure 15-11a typically fail at the points where the acceleration reverses sign, due to many cycles of crossover shock. Note also that the roller follower has to reverse direction every time it crosses over to the other side of the groove. This causes significant follower slip and high wear on the follower compared to an open, force-closed cam where the follower will have less than 1% slip.

Because there are two cam surfaces to machine and because the cam track, or groove, must be cut and ground to high precision to control the clearance, form-closed cams tend to be more expensive to manufacture than force-closed cams. Track cams usually must be ground after heat treatment to correct the distortion of the groove resulting from the high temperatures. Grinding significantly increases cost. Many force-